Cross-Platform Calibration of Whole-Genome Sequencing Genotype Quality Control Parameters Using Array-Based Genotypes

A Multi-Objective Framework for Pareto-Optimal Calibration and Knee-Point Selection



`tuning.concordance`

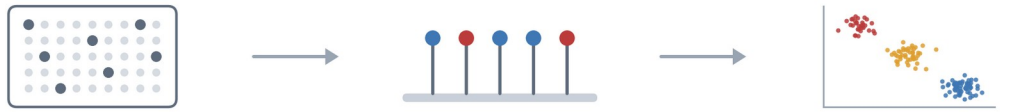
ZHAO TIE

2026-09-16

WGS Detects More Variants and Gives Direct Read-Level Evidence

SNP array

a Workflow · how each platform produces a genotype

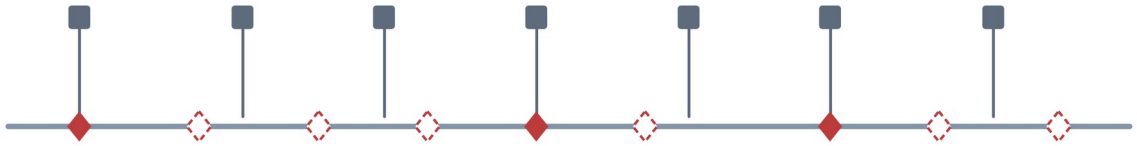


Probe design
fixed, predefined loci

Hybridization
fluorescent A / B signal

Cluster calling
genotype from intensity

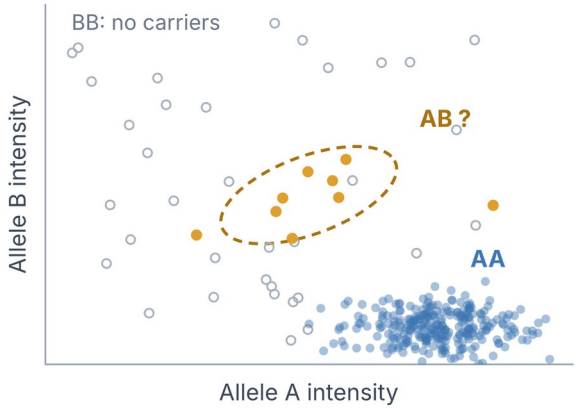
b Variant discovery · same genomic region, same true variants



3 / 9 variant sites assayed
only predefined probe loci are measured

detected · missed · probe

c Rare-variant genotype · evidence for one carrier of a low-frequency variant

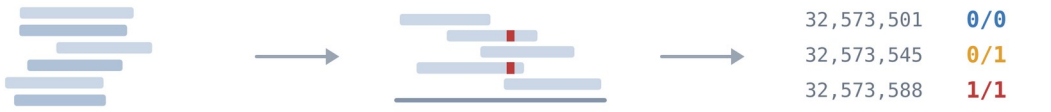


- 0/0 (AA)
- 0/1 carrier
- no-call / noise

Genotype inferred from intensity clusters

Rare-allele carriers are sparse, so the AB cluster is ill-defined and calls are uncertain.

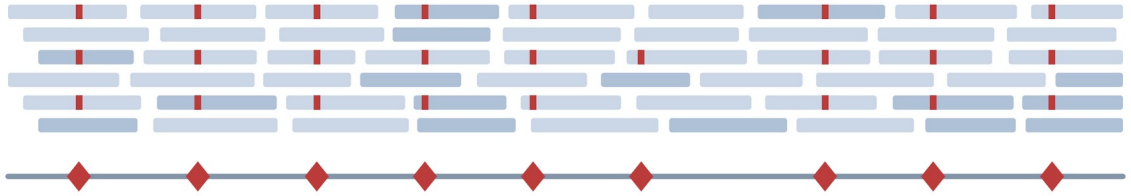
Whole-genome sequencing (WGS)



Sequencing
short reads, whole genome

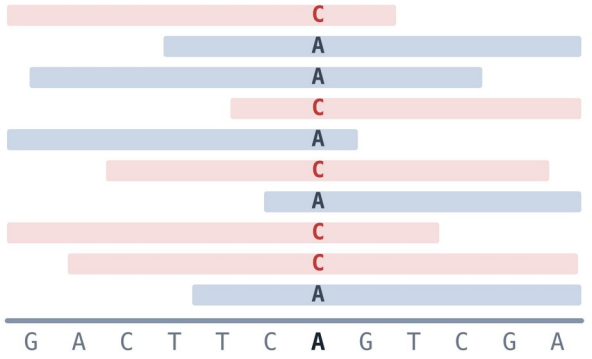
Alignment
reads mapped to reference

Variant calling
genotype from read counts



9 / 9 variant sites observed
reads cover contiguous sequence

detected · read



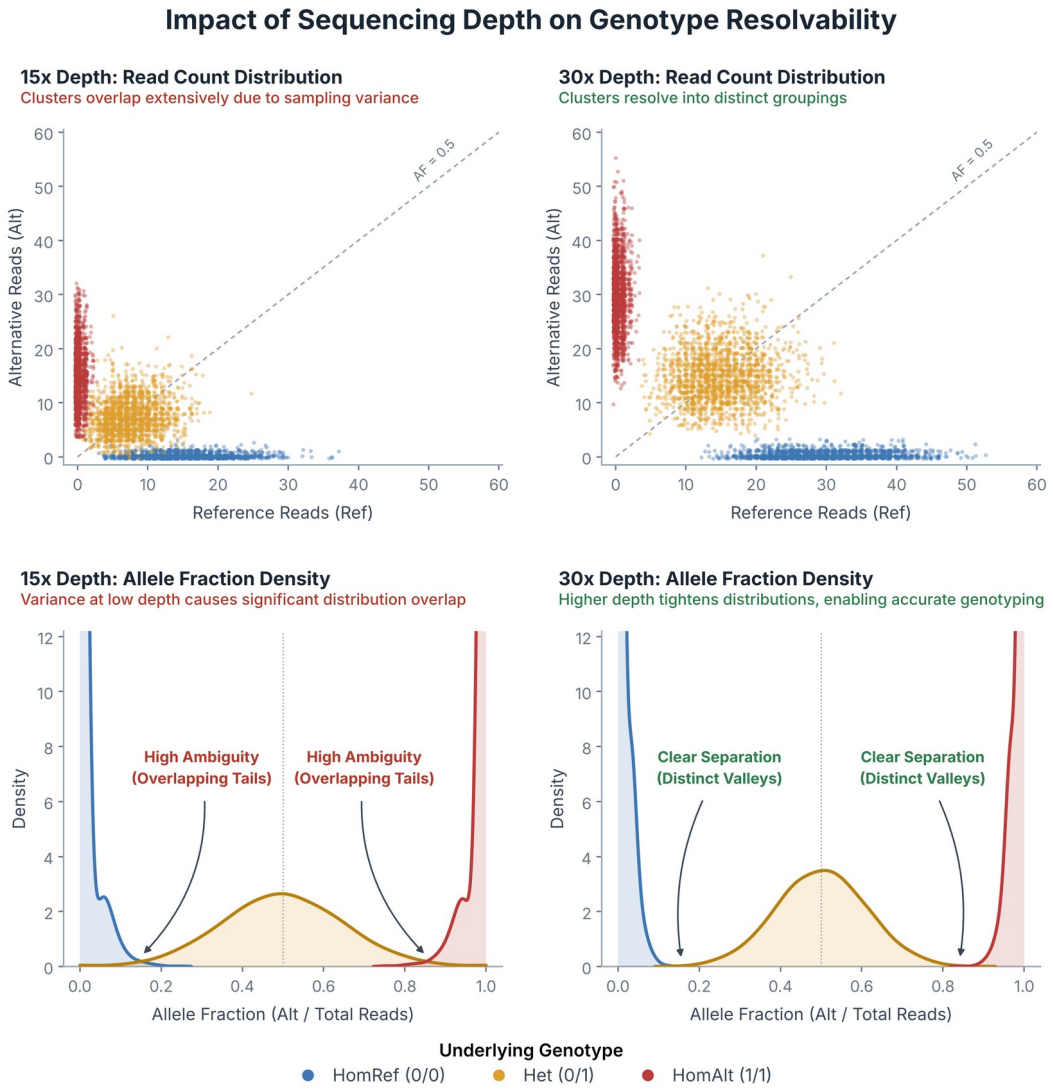
5 alt / 10 reads
reads covering the variant site



Genotype read directly from allele counts

Alt-allele reads are observed at the site, giving a 0/1 call with direct read-level evidence.

Genotype Resolution in WGS Depends on Sequencing Depth



Although WGS provides broader variant discovery and more direct read-level evidence than array genotyping, **genotype inference in WGS is inherently depth-dependent**.

Upper panels: reference vs alternate read counts

- Lower depth gives more **diffuse, overlapping** genotype clusters because of increased sampling variance
- Higher depth gives more **compact, distinctly separated** genotype groups

Lower panels: allele-fraction space

- Lower depth broadens the distributions and increases overlap between genotype classes
- Higher depth narrows the distributions and produces clearer valleys between states, improving genotype discrimination

Implication for our data

- The ability to resolve genotypes relies directly on depth; when depth varies across samples, genotype certainty also varies
- This can cause systematic differences in **missingness, calling uncertainty and error patterns**, leading to technical noise, imbalance and bias downstream, which matters in our mixed 15x/30x cohort

Most cases are sequenced at 30x and all controls at 15x, so depth-driven QC differences can bias case/control comparisons.

Study Population

			Case (CTEPH)	Ctrl (AGP3K)
Sample Information	Sample Size		457	3,135
	Age at DNA Collection[a]		64.75 ± 12.16	59.90 ± 14.07
	Sex (Male/Female)		100/357	1,243/1,892
	Collection Facility		Okayama Medical Center (n=451) Hokkaido University Hospital (n=2) Osaka Metropolitan University Hospital (n=3) Yamaguchi University Hospital (n=1)	
Clinical Information	DVT Presence (Yes/No/Unknown)[b]		Earlier study: 103/0/138 (n=241) RADDAR-J: 64/151/1 (n=216)	
	Max mPAP (mmHg)[c]		42.14 ± 11.20	
Genotyping Information	WGS	Sample Size	457	3,135
		Platform	DNBSeq-T7 30x (n=331) NovaSeq 30x (n=52) DNBseq-G400RS 15x (n=31) DNBseq-G400RS 30x (n=43)	HiSeqX 15x (n=3,135)
		Operator	Macrogen Japan (n=383) CGM (n=74)	Macrogen Korea (n=3,135)
	Array	Sample Size	321	1,442
		Platform	Illumina ASA24v1 (n=321)	Illumina ASA24v1 (n=7) Illumina JSA24v1 (n=1,433) Illumina ASA24v1 & JSA24v1 (n=2)[d]

a: Not recorded for 68 CTEPH and 66 AGP3K samples.

b: DVT, deep vein thrombosis; Unknown, not recorded.

c: mPAP, mean pulmonary arterial pressure; highest value recorded per patient, available for all CTEPH samples.

d: Two samples are genotyped on both Illumina ASA24v1 and JSA24v1.

Three Call-Level Parameters: DP, GQ and AF

To calibrate WGS genotype calls, we focused on three core call-level parameters: depth (DP), genotype quality (GQ) and allele fraction (AF).

Sequencing depth (DP)

Total reads covering a position: the sum of reference- and alternate-supporting allele depths.

- Higher DP usually gives more reliable genotypes
- Low DP increases random sampling error, which can cause unstable calls or data gaps

Genotype quality (GQ)

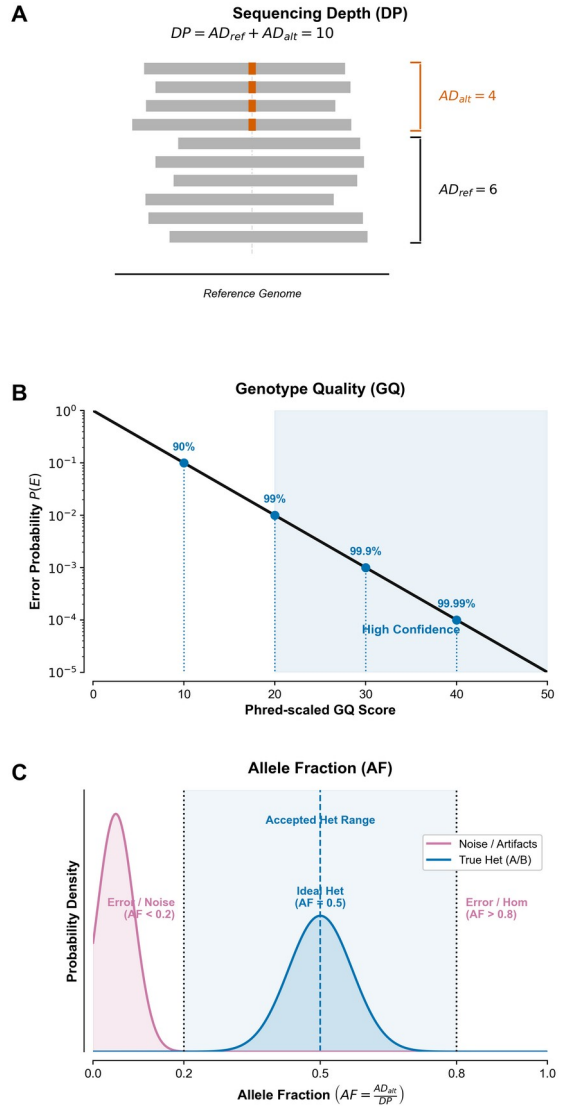
Confidence of the assigned genotype, from the PL (Phred-scaled likelihood) difference between the most and second-most likely genotypes.

- Higher GQ means lower error probability
  - GQ 10 / 20 / 30 / 40 ≈ 90% / 99% / 99.9% / 99.99% call accuracy
- A practical threshold for filtering uncertain calls

Allele fraction (AF)

Proportion of reads supporting the alternate allele. A true heterozygote should sit near 0.5, but real data deviate because of:

  - Allelic imbalance during sequencing
  - Reference bias in read alignment
  - Allele-specific mapping quality differences
- AF far outside the heterozygous range more likely reflects noise; bounding AF reduces false heterozygous calls



Each parameter removes a different kind of unreliable call, so we tune DP, GQ and AF together.

Array QC Keeps 1,750 Samples and 409,088 Variants

Phase	Exclusion Criteria	Retained	Excluded	
			Samples (Case/Ctrl)	Variants
		illuminaASA24v1: Samples: 330 (CTEPH/AGP3K=321/9); Variants: 633,357 illuminaJSA24v1: Samples: 1,435 (CTEPH/AGP3K=0/1,435); Variants: 691,920		
Sample QC	SMISS ≥ 0.01	illuminaASA24v1: Samples: 316 (CTEPH/AGP3K=309/7); Variants: 633,357 illuminaJSA24v1: Samples: 1,434 (CTEPH/AGP3K=0/1,434); Variants: 691,920	15 (12/3)[a]	
Merge	Non-shared Variants	Merged: Samples: 1,750 (CTEPH/AGP3K=309/1,441); Variants: 578,970		
Variant QC	Multiallelic Variants	Merged: Samples: 1,750 (CTEPH/AGP3K=309/1,441); Variants: 574,240		4,730
	VMISS ≥ 0.01	Variants: 565,224		9,016
	Non-autosomal Variants	Variants: 535,182		30,042
	MAF < 0.01	Variants: 409,321		125,861
	Ctrl HWE P < 1e-06 | Case HWE P < 1e-10	Variants: 409,088		233

a: **Two samples** are duplicated; since SMISS, only the samples in illuminaJSA24v1 are retained.

SMISS: Sample Missing Rate
VMISS: Variant Missing Rate
MAF: Minor Allele Frequency
HWE P: P-value of Hardy-Weinberg Equilibrium

ToMMo Frequency Filter Keeps 385,609 High-Confidence Variants

Why filter against ToMMo

- Array QC removes genotyping noise; ToMMo, a larger Japanese reference, flags low-confidence and population-specific variants
- Keep variants whose control AAF − ToMMo AAF falls in the central **95%** range (−0.023 to 0.023)
- Result: **385,609** of 405,904 valid variants kept as high-confidence anchors

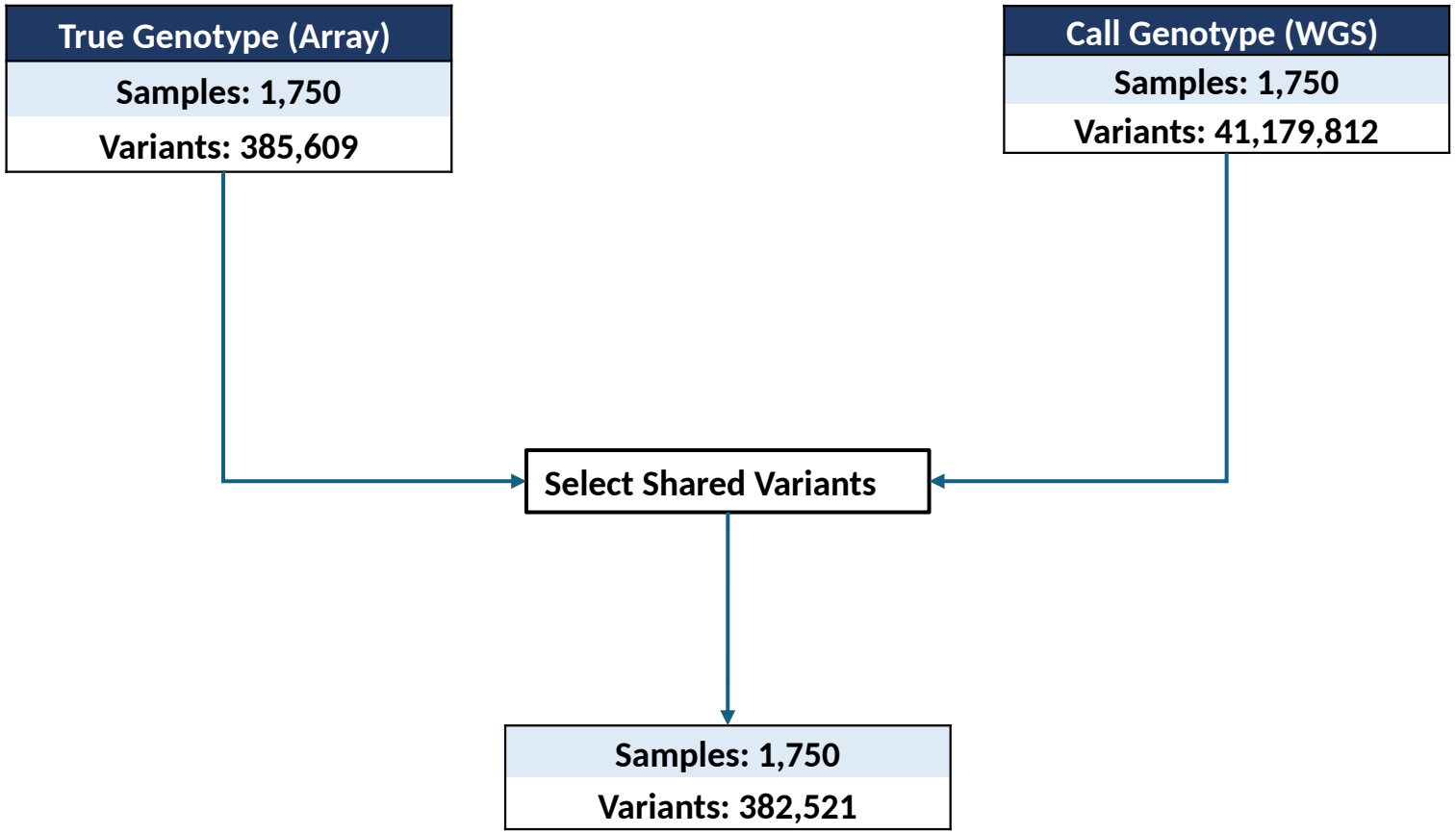
```
============================================================
        ToMMo 95% Central Range Filter Report
============================================================
Date: 2025-12-29 00:21:28
Input File: cteph_agp3k.array.sqc.vqc.variant_qc_summary.variant_qc_with_tommo.tsv.gz
Output Prefix: cteph_agp3k

------------------------------------------------------------
Filtering Statistics
------------------------------------------------------------
1. Total Variants Loaded:                  409,088
2. Variants with TOMMO_FILTER='PASS':      405,904 (99.22%)
3. Valid AAF Data (Non-NaN):               405,904

------------------------------------------------------------
95% Central Range Calculation (DIFF = AAF_CTRL - TOMMO_AAF)
------------------------------------------------------------
Lower Bound (2.5th percentile):  -0.023288
Upper Bound (97.5th percentile): 0.022885

------------------------------------------------------------
Final Filtering Results
------------------------------------------------------------
Retained Variants (Inside Range):          385,609 (95.00% of valid PASS)
Excluded Variants (Outside Range):          20,295
```

382,521 Variants Shared by Array and WGS Form the Test Set



Truth set: array genotypes

- 1,750 samples after array QC
- 385,609 variants after the ToMMo filter

Test set: WGS calls

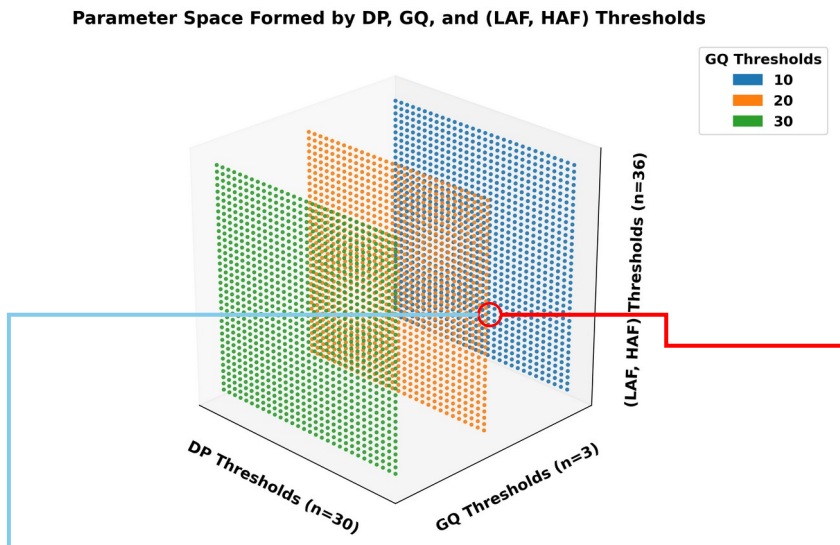
- Same 1,750 samples
- 41,179,812 variants called

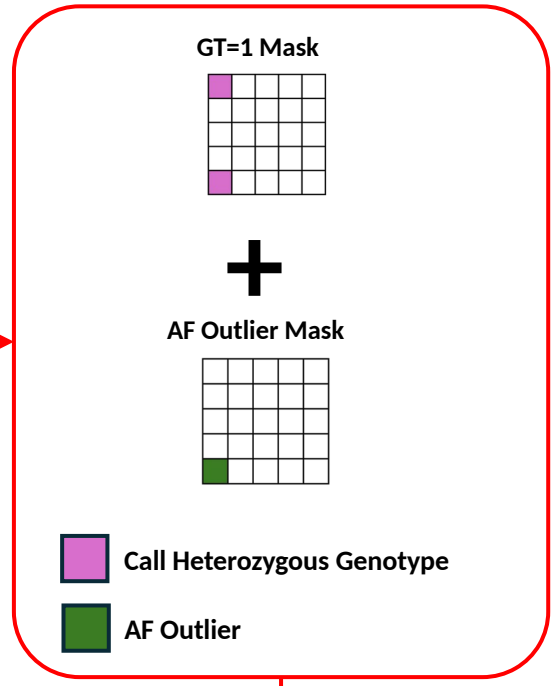
Comparison set

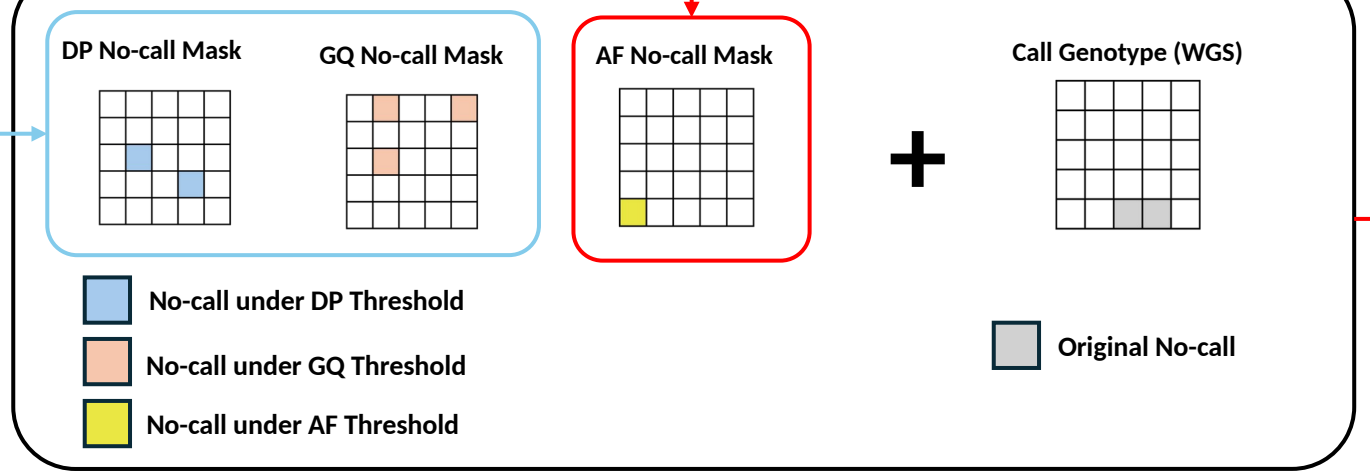
- **382,521** variants present on both platforms

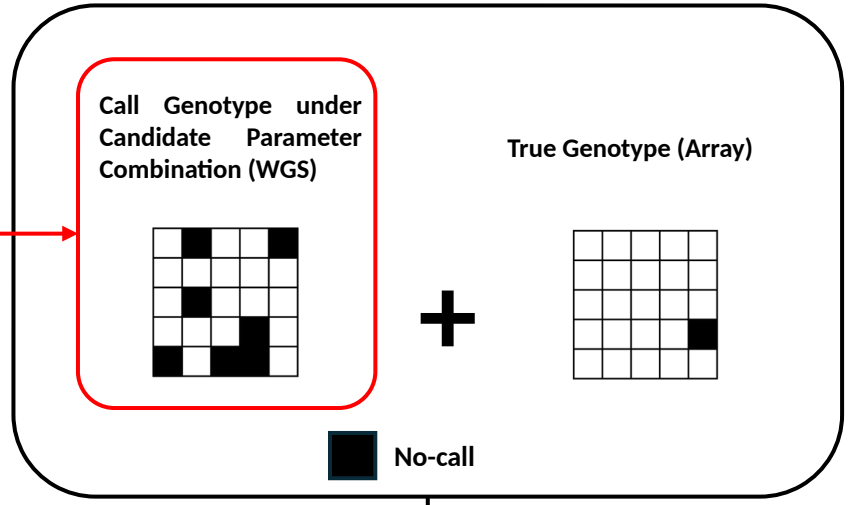
Every metric that follows compares WGS and array genotypes at the same 382,521 sites in the same 1,750 samples.

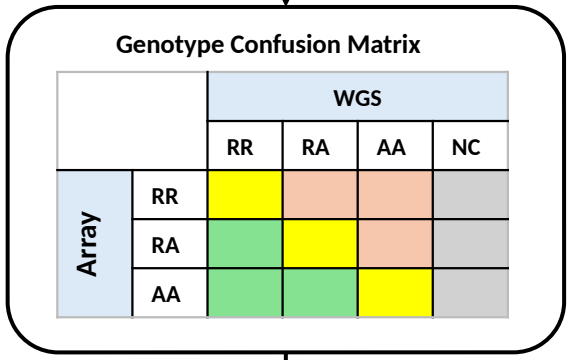
Each DP/GQ/AF Combination Yields One Confusion Matrix











Genotype Consistency Metrics

Genotype Confusion Matrix: WGS Calls Against Array Truth

		WGS			
		Ref / Ref	Ref / Alt	Alt / Alt	No-call
Array	Ref / Ref				
	Ref / Alt				
	Alt / Alt				

FP: False Positive

TP: True Positive

TN: True Negative

FN: False Negative

Missing

Reading each cell

- **TN**: array Ref/Ref and WGS Ref/Ref
- **TP**: WGS matches a non-reference array genotype (Ref/Alt or Alt/Alt)
- **FP**: WGS calls more alternate alleles than the array
- **FN**: WGS calls fewer alternate alleles than the array
- **Missing**: WGS No-call, whatever the array genotype

How cells are counted

- Rows: array genotype (truth); columns: WGS genotype under the tested DP, GQ and AF thresholds
- In WGS data, **No-calls include both original missing values** and genotypes reclassified as missing under each parameter threshold
- The confusion matrix and consistency metrics follow Kishikawa et al. (2019) [1]

Each threshold combination produces one 3 × 4 matrix; every consistency metric is a ratio of its cells.

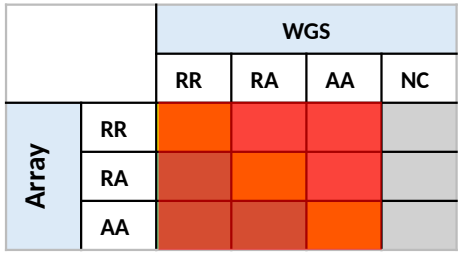
[1] Kishikawa, T., Momozawa, Y., Ozeki, T., Mushiroda, T., Inohara, H., Kamatani, Y., Kubo, M., & Okada, Y. (2019). Empirical evaluation of variant calling accuracy using ultra-deep whole-genome sequencing data. Scientific Reports, 9(1), 1784.

Consistency Metrics (1): Concordance and Miss Rate

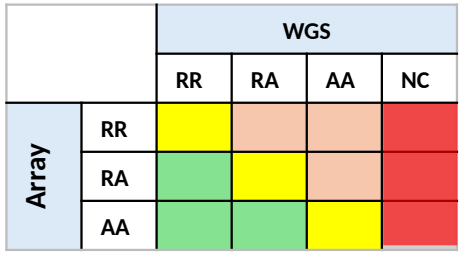
Each metric is the sum of the highlighted cells in the numerator matrix divided by the sum in the denominator matrix.

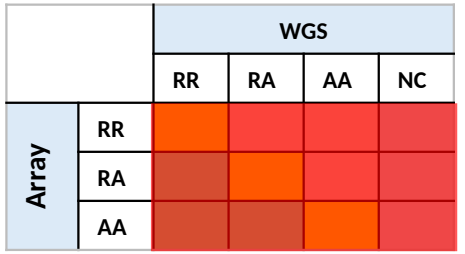
Genotype Concordance =





Genotype Miss Rate =



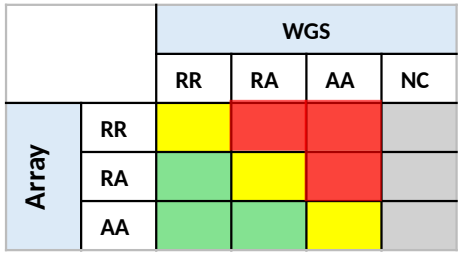


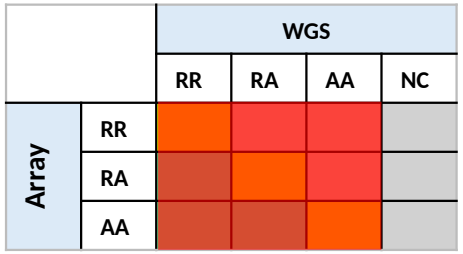
Concordance and miss rate come from the same matrix, so any threshold change moves both at once.

Consistency Metrics (2): False Positive and False Negative Rates

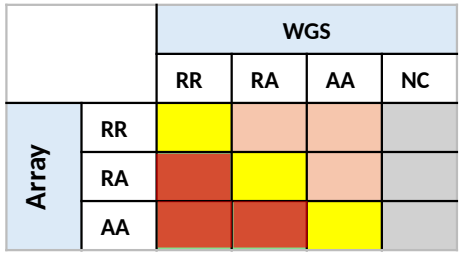
Same reading as the previous slide: highlighted numerator cells over highlighted denominator cells.

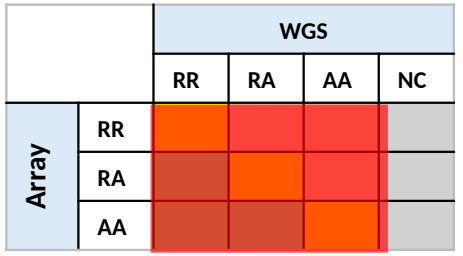
False Positive Rate =





False Negative Rate =

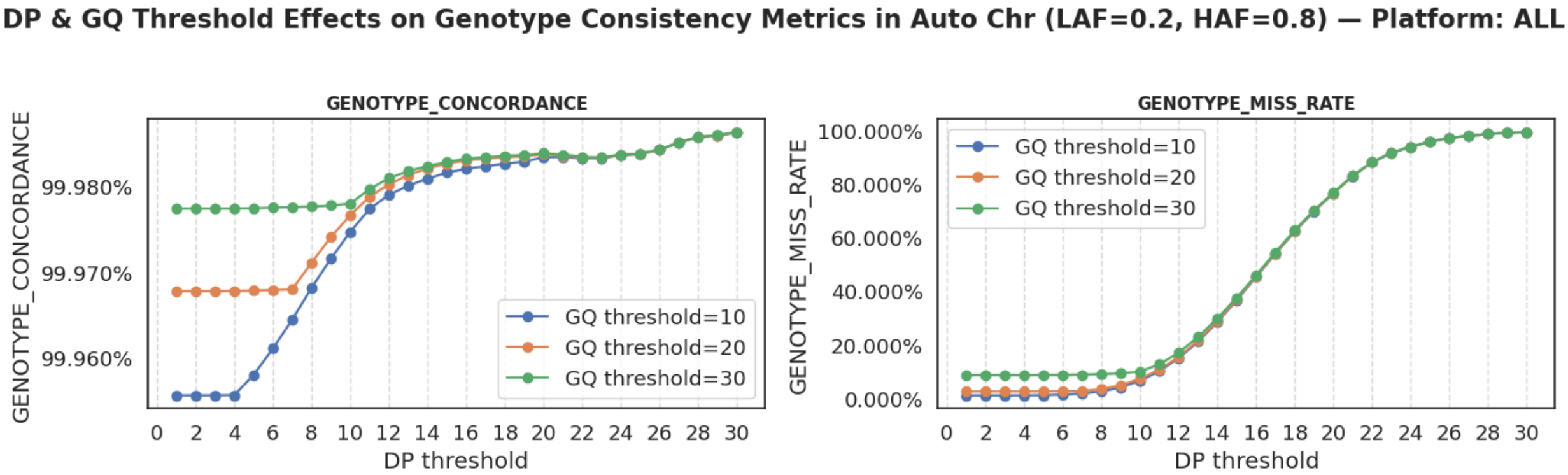




False positive and false negative rates split discordant calls by direction, which concordance alone cannot show.

Higher DP and GQ Raise Concordance; Miss Rate Climbs Past DP 10

AF range fixed at (0.20, 0.80); autosomes, all platforms; DP threshold 1–30, GQ threshold 10 / 20 / 30.



Genotype concordance

- The concordance rate **steadily increases** with higher DP thresholds
- Higher GQ thresholds give a **better baseline** concordance
- All settings converge to nearly 100% concordance at high DP

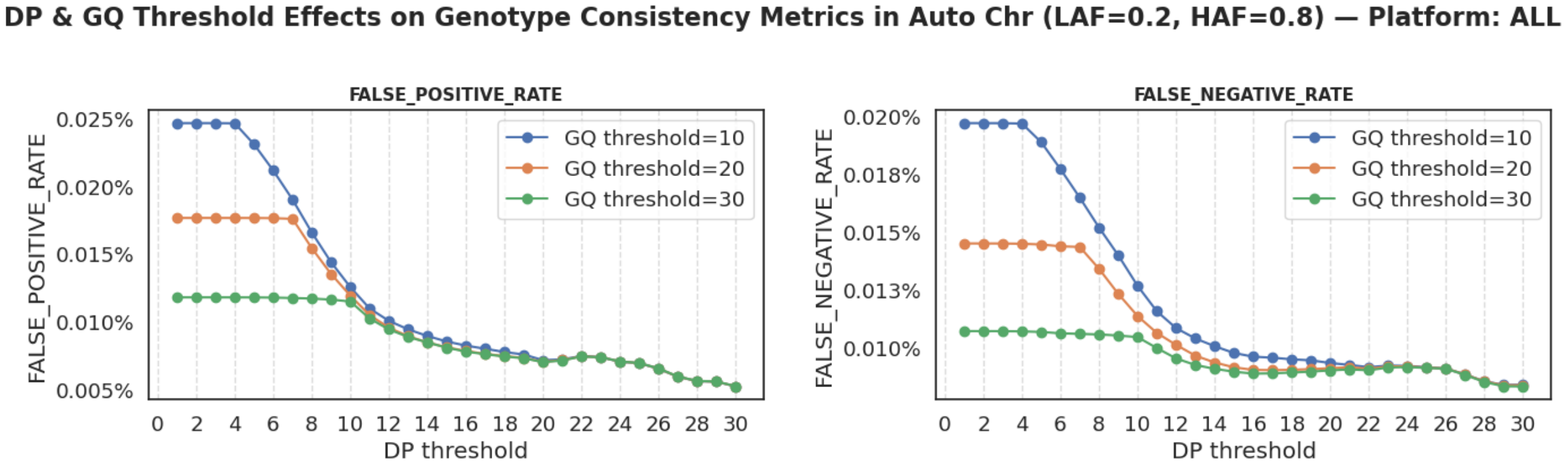
Genotype miss rate

- Remains **low** at low DP thresholds
- Begins to **rise once the DP threshold exceeds ~10**, eventually reaching about 100% by DP 30

Above DP ≈ 10, further concordance gains are small while the miss rate rises fast.

Higher DP and GQ Lower False Positive and False Negative Rates

AF range fixed at (0.20, 0.80); autosomes, all platforms; DP threshold 1–30, GQ threshold 10 / 20 / 30.



False positive rate

- Initial values **decrease as the GQ threshold increases**
- **Continuously decreases** with increasing DP threshold

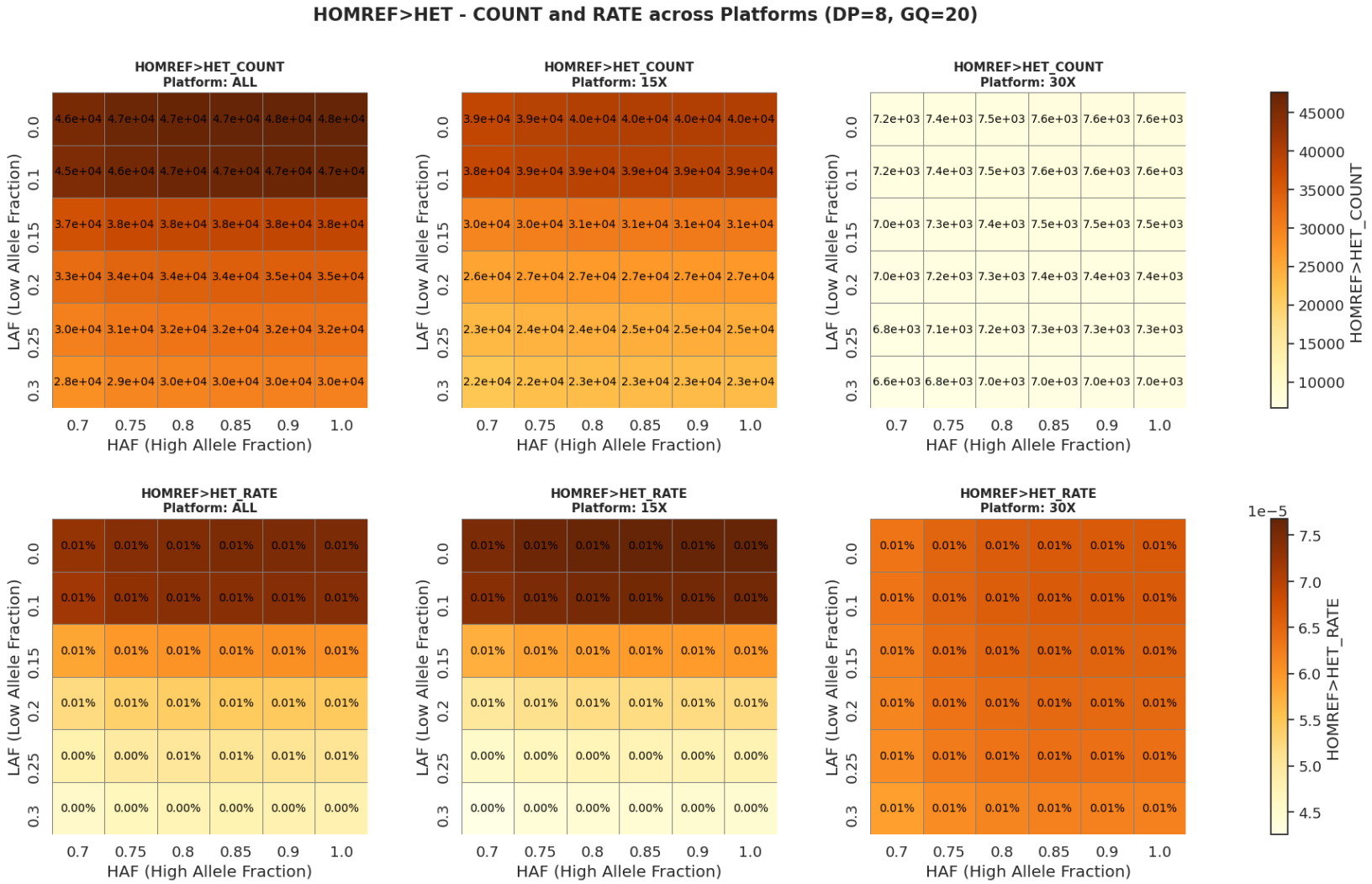
False negative rate

- Shows **similar trends** to the false positive rate: a lower starting value at higher GQ, then a steady decline as the DP threshold rises

Both error rates fall with stricter DP and GQ, and the GQ settings converge above DP ≈ 10.

Tighter AF Bounds Cut HOMREF>HET Errors, Mostly at 15x

DP threshold 8, GQ threshold 20; LAF (low allele-fraction bound) by HAF (high bound); top: counts, bottom: rates.



The heatmaps show how the HOMREF>HET conversion rate changes as the AF bounds vary.

Effect of the AF bounds

- **Stricter AF settings** (higher LAF, lower HAF) notably decrease the conversion rate

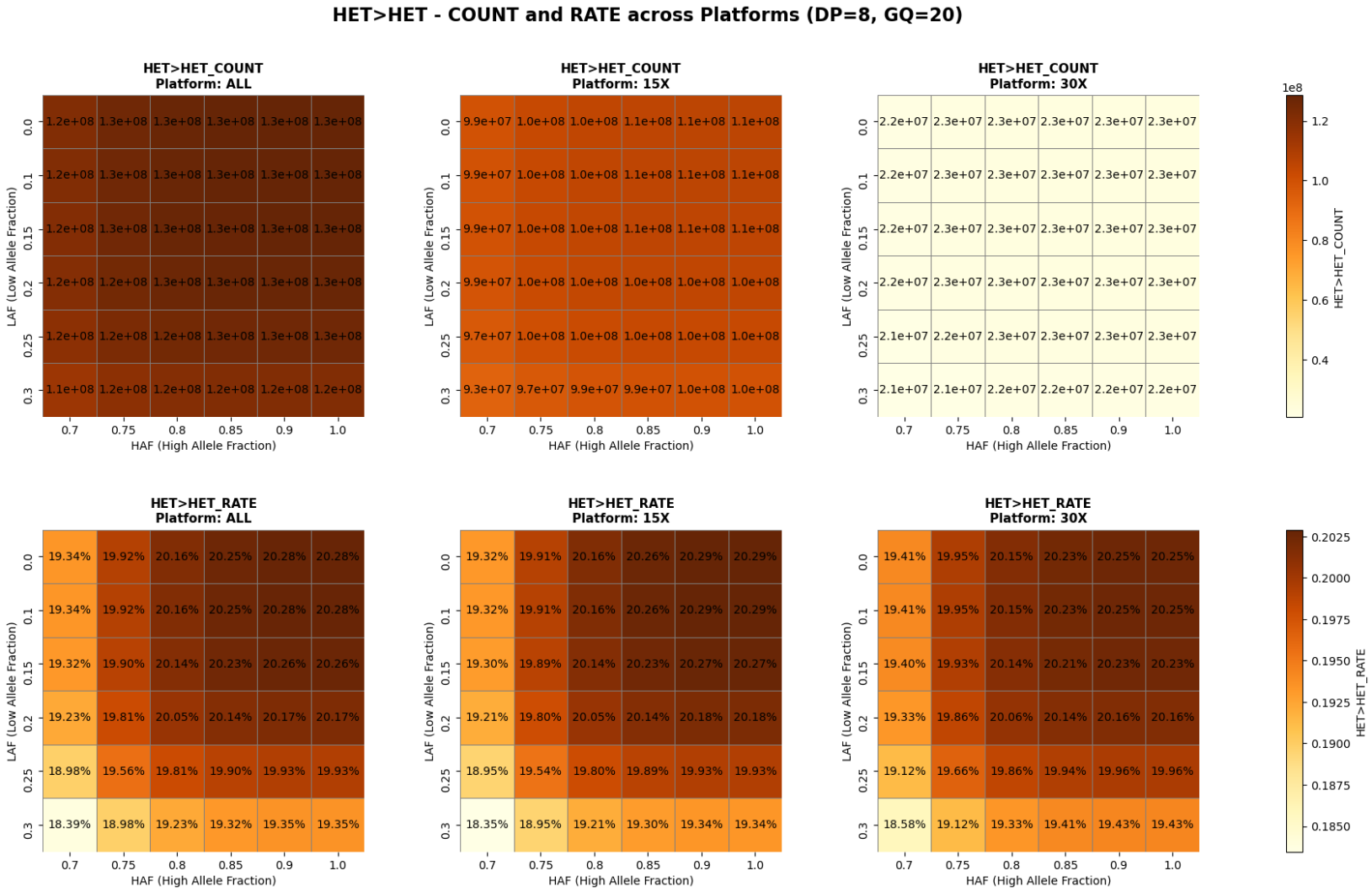
Platform differences

- **30x** samples genotype more consistently as the thresholds change
- **15x** samples show higher misclassification under more lenient thresholds
- Tighter AF bounds **reduce the difference** between 15x and 30x

Tight AF bounds matter most for 15x samples, where loose bounds let reference homozygotes through as heterozygotes.

Tighter AF Bounds Also Lose True HET>HET Calls

DP threshold 8, GQ threshold 20; LAF (low allele-fraction bound) by HAF (high bound); top: counts, bottom: rates.



The heatmaps show how the rate of correctly identified HET>HET calls changes as the AF bounds vary.

Effect of the AF bounds

- Implementing **stricter AF thresholds** (higher LAF, lower HAF) leads to **fewer correctly identified HET>HET cases**

Cost of tight bounds

- The same settings that remove HOMREF>HET errors on the previous slide also discard true heterozygous calls

AF bounds trade false heterozygous calls against lost true ones, so they cannot simply be set as tight as possible.

Stricter Thresholds Trade Data Retention for Accuracy

What the threshold scans show

- Stricter thresholds for DP, GQ and AF **improve genotype call accuracy**: higher concordance, lower false positive and false negative rates
- They also **filter more data**, which raises the genotype miss rate

What we need

- In practical use, we must find a **balance between accuracy and data retention**
- This is a **multi-objective optimization** problem, introduced on the next slides



No single threshold maximizes accuracy and retention together; the choice is a trade-off.

Threshold Choice Is a Two-Objective Problem

Multi-objective optimization identifies the best solutions when several goals compete. Unlike single-objective problems, which usually have one clear solution, multi-objective problems have a **set of optimal solutions, the Pareto solutions**, where improving one objective typically requires sacrificing another.

In genotype QC we must balance two metrics: **genotype concordance** and **genotype miss rate**.

Two goals in genotype QC

- **Maximize** genotype concordance
- **Minimize** genotype miss rate

Why it is a classic trade-off

- **Trade-offs are unavoidable:** improving concordance often increases the miss rate
- **The metrics are interconnected:** adjusting QC parameters changes concordance and miss rate at the same time
- **No perfect solution exists:** both cannot be optimized at once

Our objectives

- Identify a set of optimal QC parameters (the **Pareto front**)
- Maximize the retention of valid data while ensuring data quality
- Provide clear trade-off information across parameter configurations so we can select the most suitable one

We need a rule that picks one point from the Pareto front, not a single best value for each metric.

Pareto Front: Settings No Other Setting Beats on Both Goals

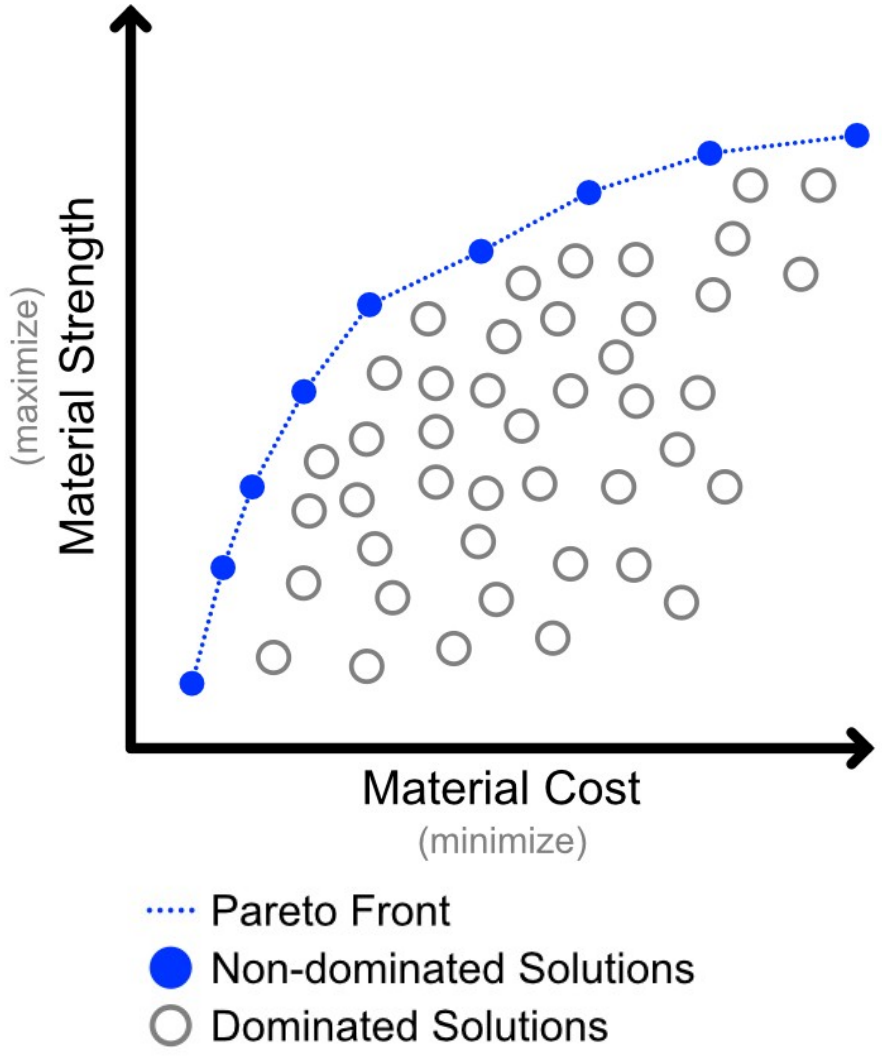
Pareto optimization is a fundamental technique for multi-objective optimization. It relies on three concepts:

Key concepts

- **Dominated solution (gray points):** performs worse on both objectives than some point on the Pareto front
- **Non-dominated solution (blue points):** no other solution is better on all objectives
- **Pareto front (blue curve):** the set of all non-dominated solutions; on this curve, improving one objective inevitably compromises the other

Reading the diagram

- The diagram shows a bi-objective problem; each point on the Pareto front is an optimal balance between the two objectives



Only points on the Pareto front are worth considering; every other point can be improved on both goals.

Applied to QC: Miss Rate Against Concordance

In our example, the x-axis is **genotype miss rate** and the y-axis is **genotype concordance**; each point is the result of one parameter setting.

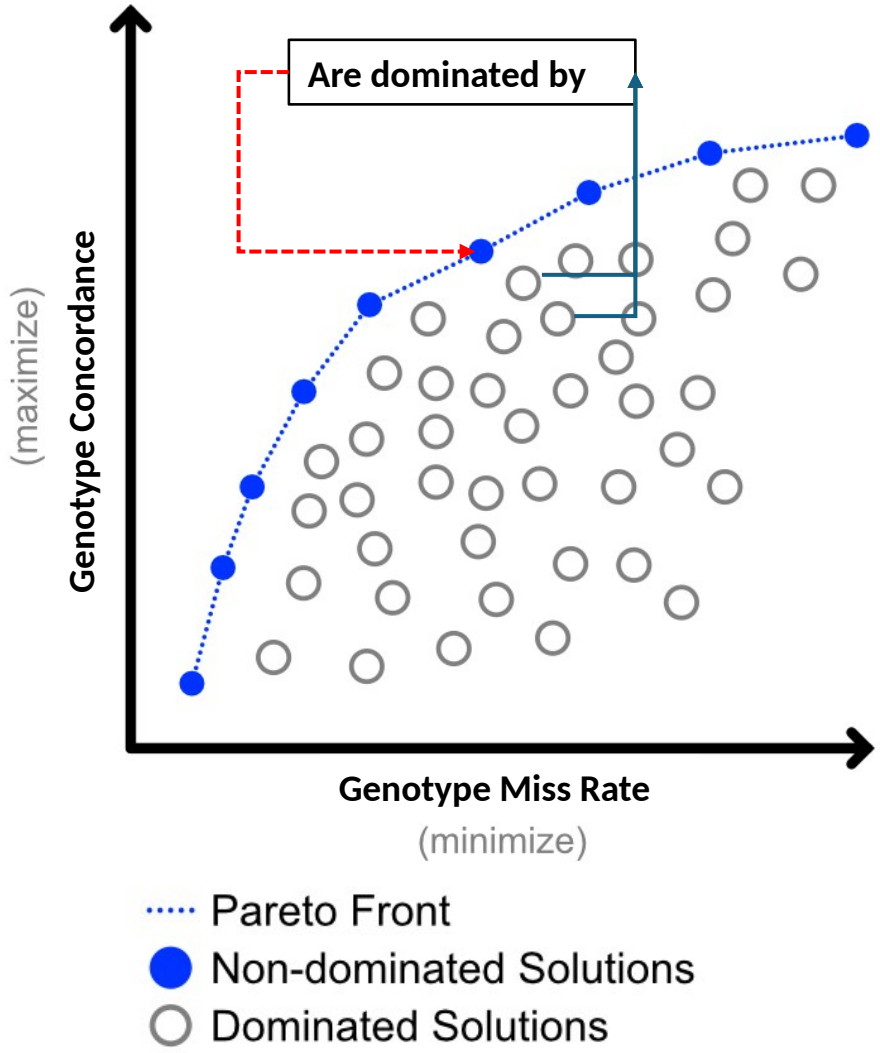
What the points mean

- **Gray points (dominated):** some parameter combination achieves both a lower miss rate and higher concordance
- **Blue points (non-dominated, on the Pareto front):** the best trade-offs; no other setting is better on both metrics
- **Blue curve (Pareto front):** moving along it, increasing concordance requires accepting a higher miss rate

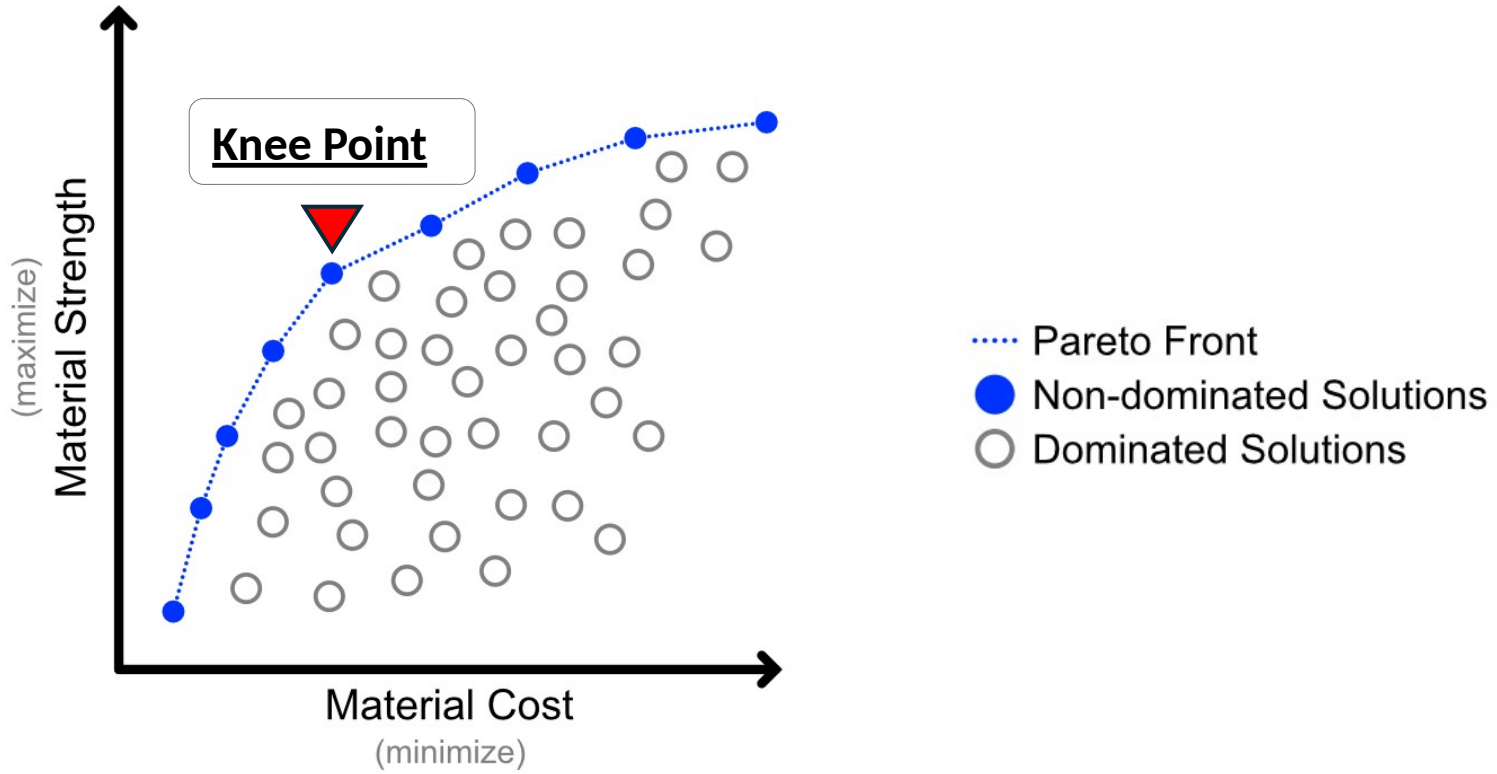
How we use it

- This lets us focus only on the optimal parameter combinations: the non-dominated solutions on the Pareto front

We only choose among Pareto-front settings; the dominated settings are discarded.



We Pick the Knee Point on the Pareto Front



Why the knee point

- In practice we select QC parameters from the Pareto front according to research needs
- The knee point gives the **best balance among the metrics**; optimizing beyond it significantly harms the other metric

Fine-tuning from the knee

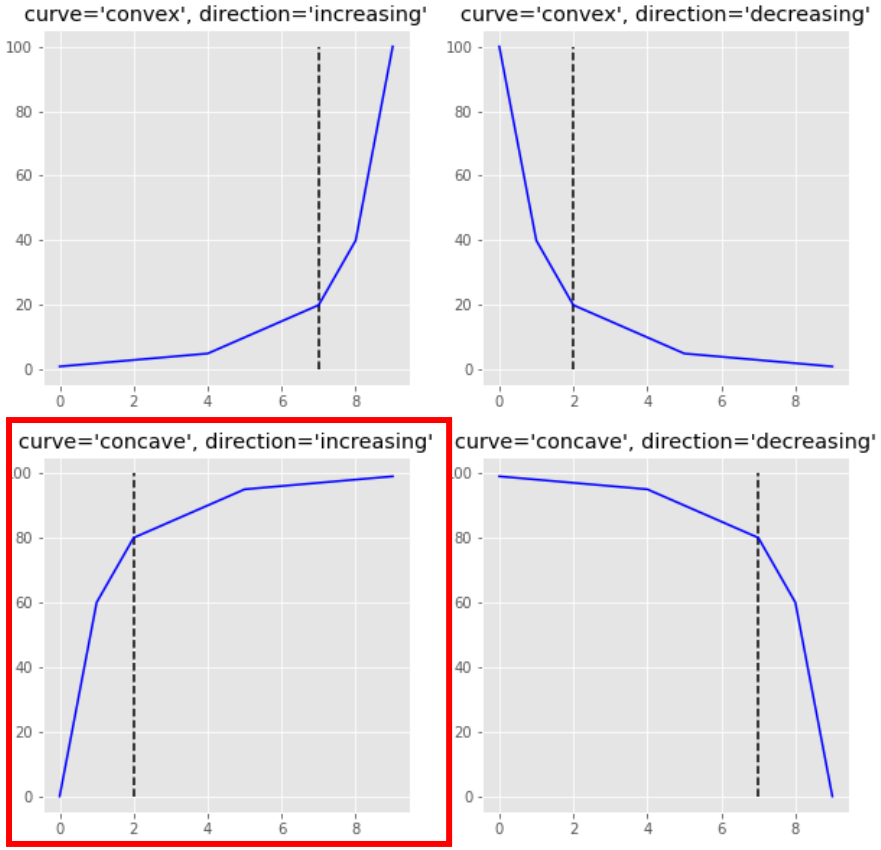
- Shift toward **data completeness** for large-scale analyses
- Shift toward **genotype accuracy** for studies that need high confidence

Next challenges

- **Identify the knee point:** the result depends on the scale and dimensions of both x and y
- **Weighted methodology:** let us prioritize one objective when optimizing the parameters

We take the knee point as the default choice and next need a method to find it and to weight it.

Kneedle Finds the Knee Point of a Monotonic Curve



Finding a "Kneedle" in a Haystack: Detecting Knee Points in System Behavior

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***Abstract*—Computer systems often reach a point at which the relative cost to increase some tunable parameter is no longer worth the corresponding performance benefit. These "knees" typically represent beneficial points that system designers have long selected to best balance inherent trade-offs. While prior work largely uses ad hoc, system-specific approaches to detect knees, we present Kneedle, a general approach to online and offline knee detection that is applicable to a wide range of systems. We define a knee formally for continuous functions using the mathematical concept of curvature and compare our definition against alternatives. We then evaluate Kneedle's accuracy against existing algorithms on both synthetic and real data sets, and evaluate its performance in two different applications.**

encounter knee detection problems similar to those we describe [1], [2], [3], [4], [5]. In these systems, researchers either use ad hoc or system-specific approaches to detect knees, or defer the problem to future work. While a finely-crafted system-specific approach will perform better than a general knee detection approach, a designer may not take the time to design one. Thus, our aim is not to improve or optimize a specific system or protocol, but to provide system designers a general tool for improving the parts of their system they generally do not take the time to optimize. In network protocol and system design, rules-of-thumb often

We use the Python **kneed** library to detect knee points. Its algorithm, **Kneedle** [1], identifies the knee point of a monotonic curve. The next two slides walk through the implementation on the **concave increasing** example from the original paper (red box).

Kneedle gives a reproducible rule for picking the knee point on our Pareto fronts.

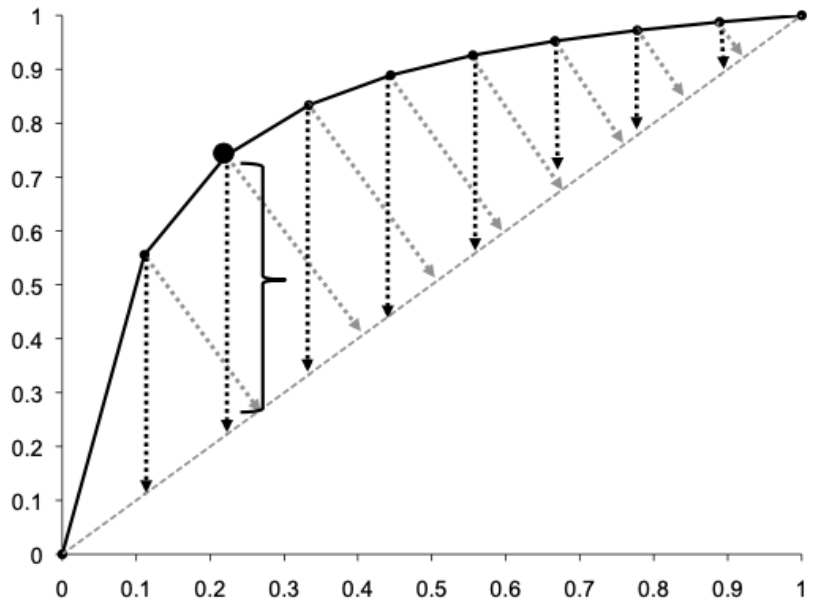
[1] Satopää, V., Albrecht, J., Irwin, D., & Raghavan, B. (2011). Finding a "Kneedle" in a haystack: Detecting knee points in system behavior. 31st International Conference on Distributed Computing Systems Workshops, 166–171. Code: github.com/arvkevi/kneed

Kneedle (1): From Input Points to the Difference Curve

1 Input curve

Monotonically increasing points; here x = miss rate and y = concordance.

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$



2 Smooth curve

Linear interpolation (scipy.interpolate.interp1d) turns the points into a continuous f(x) and damps noise.

$$f(x) = \begin{cases} y_i + \frac{y_{i+1}-y_i}{x_{i+1}-x_i}(x - x_i), & x \in [x_i, x_{i+1}] \\ \text{where } i \text{ satisfies } x_i \leq x < x_{i+1} \end{cases}$$

$$\text{and } f(x) \text{ is a monotonically increasing, concave function}$$

3 Normalize

MinMax scaling maps x and f(x) onto [0, 1], so the two axes are comparable.

$$x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad y_{\text{norm}} = \frac{f(x) - \min(f(x))}{\max(f(x)) - \min(f(x))}$$

4 Difference curve

Compare f with the line y = x (linear growth). A curve that rises fast and then flattens sits above the line early and approaches it later; D(x) measures that gap.

$$D(x) = y_{\text{norm}} - x_{\text{norm}}$$

A larger D(x) means the curve sits further above y = x: the region of greatest gain.

Kneedle (2): From Difference-Curve Peaks to the Knee Point

5 Local maxima become candidates

Each local maximum of D(x) marks a peak in deviation from y = x and is a knee candidate.

$$D(x_i) \text{ is a local maximum} \quad \text{when } D(x_i) \geq D(x_{i-1}) \text{ and } D(x_i) \geq D(x_{i+1})$$

6 Threshold

A threshold T below each maximum keeps small fluctuations from being taken as knees; S sets its sensitivity (default 1).

$$T(x) = D_{\max} - S \cdot \text{mean}(\Delta x)$$

$$D_{\max} : \text{local maximum of difference function}$$

$$\Delta x : \text{step size of normalized } x$$

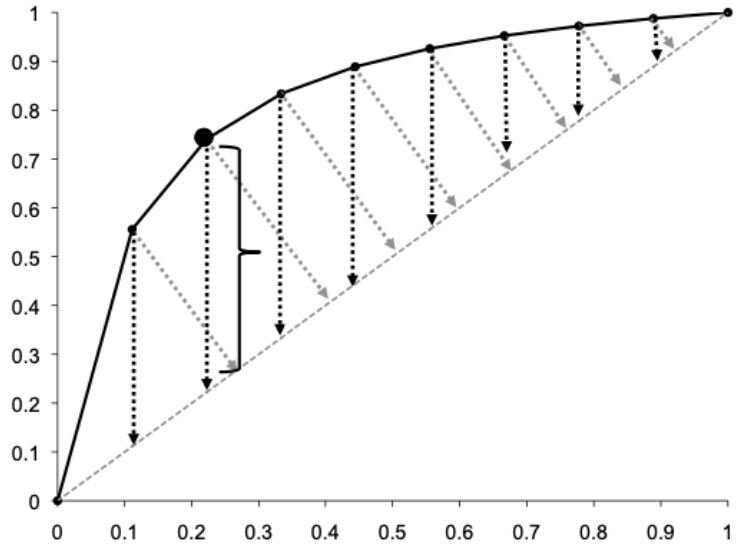
$$S : \text{sensitivity parameter determining threshold for "diminishing returns"}, default = 1$$

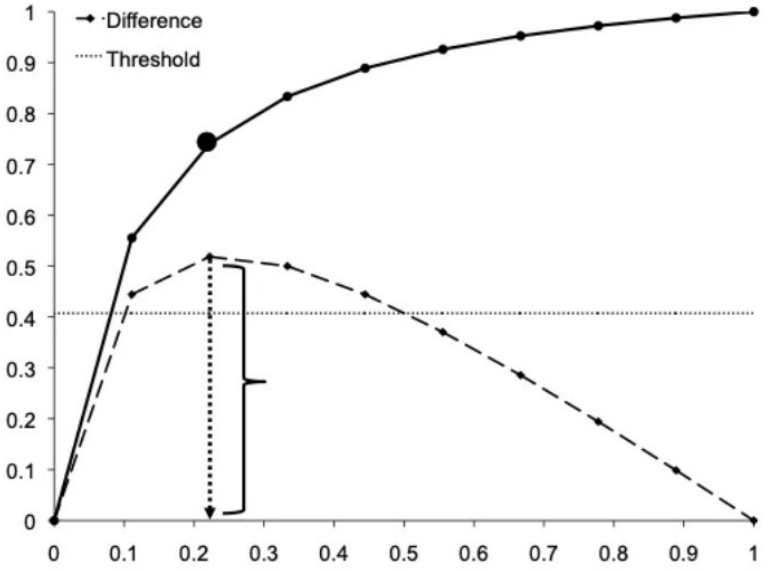
7 Detect the knee

Moving right from the first maximum, take the first point x_j that meets:

$$D(x_j) > T(x)$$

The curve has entered the diminishing-returns region, and x_j is reported as the knee point.





Kneedle returns one knee per curve, set by where gains in y stop outpacing x.

Weighted Kneedle Lets One Objective Count More

Limitation of basic Kneedle

- The basic algorithm **treats all objectives as equally important**, which limits it in multi-objective optimization
- In real-world scenarios objectives often carry different weights, for example **accepting a higher miss rate to improve concordance**, or prioritizing **fewer false positives and false negatives**

Weighted extension

- We introduce a weighted extension implemented through a **Weighted Difference Curve (right)**

Weighted difference curve

$$D_w(x) = w_y \cdot y_{\text{norm}}(x) - w_x \cdot x_{\text{norm}}(x)$$

- Miss rate (x) and concordance (y) are MinMax-normalized to [0, 1]
- The y weight scales the benefit (concordance); the x weight scales the cost (miss rate)
- W = y weight : x weight is the ratio used on the result slides

How the weight ratio moves the knee point

W < 1: favour retention

- Miss rate costs more than concordance gains
- The knee appears **earlier**: looser thresholds, more genotypes kept

W = 1: basic Kneedle

- The weighted curve reduces to the original difference curve D(x)
- Both objectives count equally

W > 1: favour concordance

- Concordance gains outweigh extra missingness
- The knee moves **right**: stricter thresholds, higher miss rate

One weight ratio turns a stated preference into a single, reproducible knee point.

Why the Weighted Difference Curve Is Valid

Mathematical: a linear transformation

- $D_w(x)$ is a linear combination of x_{norm} and y_{norm}: a projection onto one score axis, like a PCA projection
- A linear combination keeps convexity and the extreme points, so Kneedle's detection logic still holds

Logical: encodes a stated preference

- A weight slider sets how much extra cost we accept for a gain in concordance

Geometric: tilts the reference line

- The reference line's slope changes from 1 to w_y / w_x
- $w_y > w_x$: the line tilts up, more x is needed to offset a gain in y, and the knee moves **right**
- $w_x > w_y$: the line tilts down and the knee appears **earlier**

Weighted difference curve

$$D_w(x) = \underbrace{w_y \cdot y_{\text{norm}}}_{Benefit} - \underbrace{w_x \cdot x_{\text{norm}}}_{Cost}$$

Why it matters

- It changes **how the trade-off between improvement and cost is measured**, not the knee-search procedure
- Tilting the reference line makes knee detection **more sensitive or more conservative**

Low weight on concordance (W = 1/7 – 1/4)

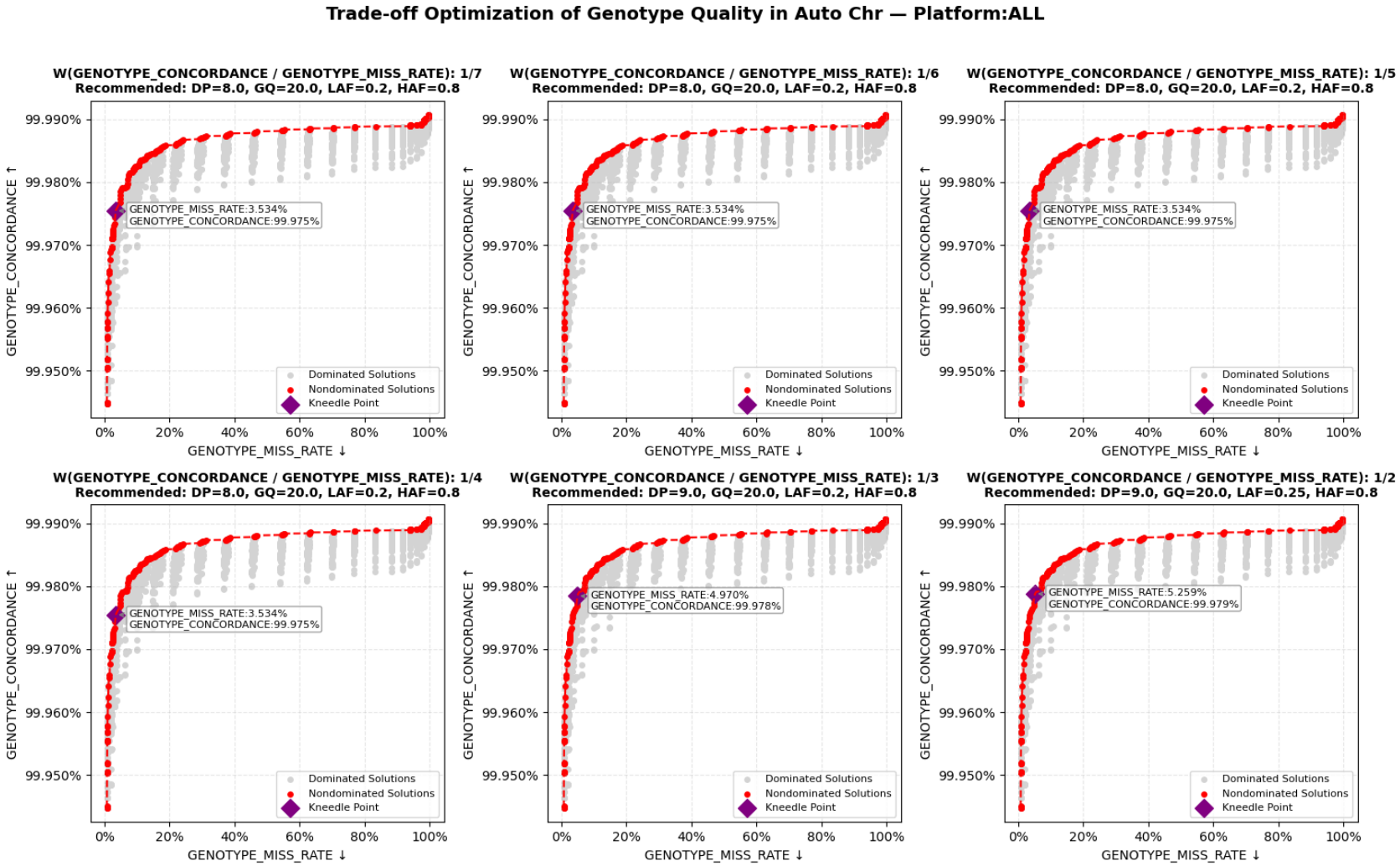
- Reference line tilts down; knee stays early at **DP 8, GQ 20, AF 0.20–0.80** (miss rate 3.53%)

High weight on concordance (W = 2 – 6)

- Reference line tilts up; knee moves right to **DP 11, GQ 30, AF 0.25–0.75** (miss rate 13.23%)

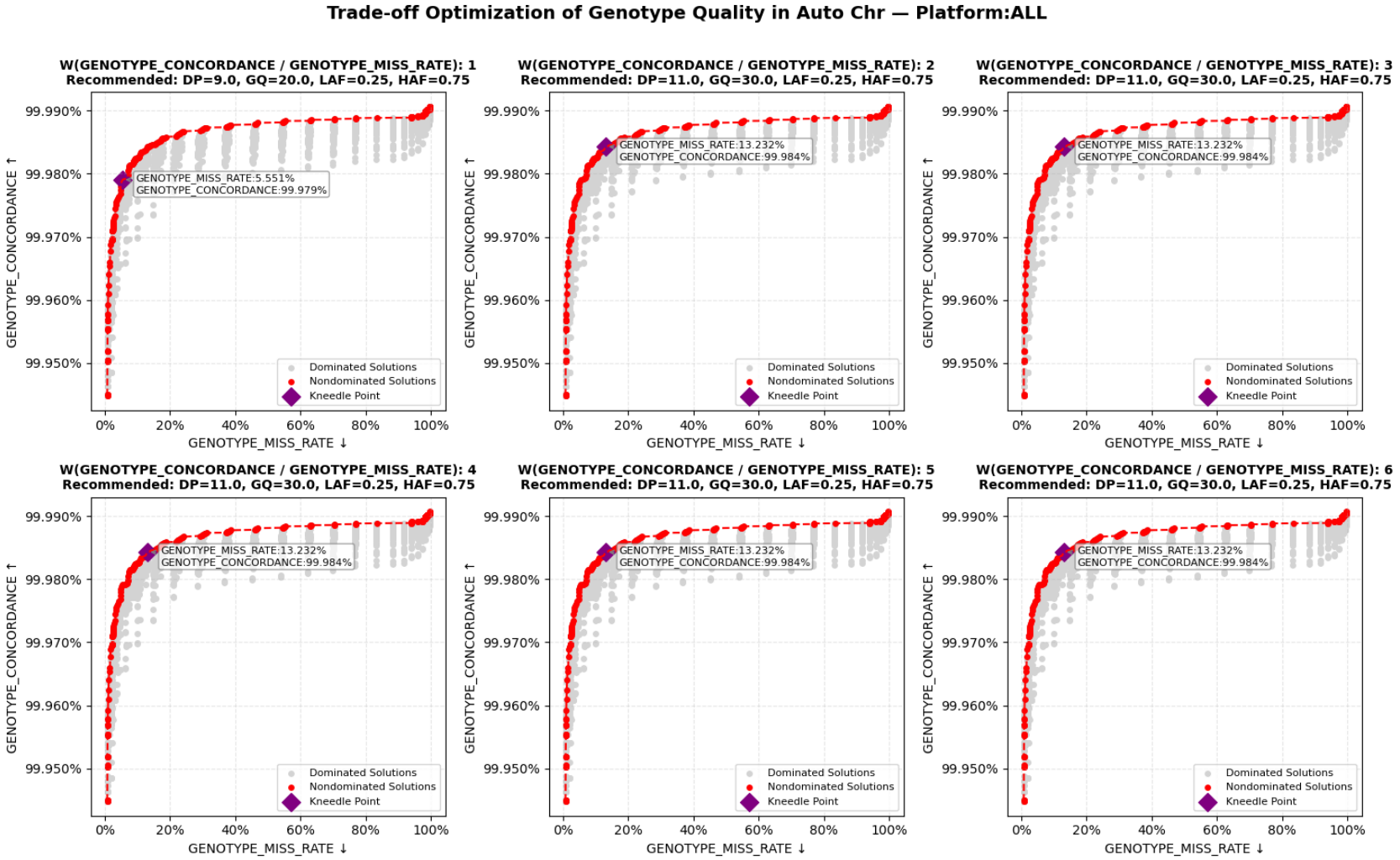
Weighting changes only the slope of the reference line, so the knee search is unchanged and the weight sets where the knee lands.

W = 1/7 to 1/2: DP 8, GQ 20 Holds Through W = 1/4



From W = 1/7 to 1/4 the knee stays at DP 8, GQ 20, LAF 0.20, HAF 0.80.

W = 1 to 6: Thresholds Tighten as Concordance Weighs More



As W rises from 1 to 6, the knee moves to stricter DP, GQ and AF thresholds.

More Weight on Concordance Raises Thresholds and Miss Rate

Knee point and metrics at each weight interval (W = weight on concordance : weight on miss rate; shaded column = recommended setting)

	W = 1/7 – 1/4	W = 1/3	W = 1/2	W = 1	W = 2 – 6
DP threshold	**8**	9	9	9	11
GQ threshold	**20**	20	20	20	30
LAF	**0.20**	0.20	0.25	0.25	0.25
HAF	**0.80**	0.80	0.80	0.75	0.75
Concordance	**99.975%**	99.978%	99.979%	99.979%	99.984%
Miss rate	**3.534%**	4.370%	5.259%	5.551%	13.232%

Weights and recommended parameters

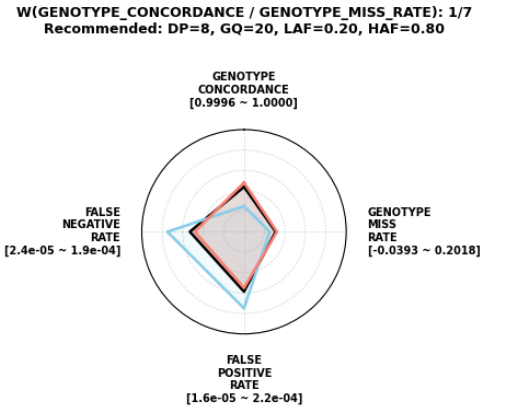
- As W shifts from 1:7 to 6:1, the selection **prioritizes concordance more**
- **DP** rises from 8 to 11 and **GQ** from 20 to 30: stricter depth and quality criteria
- **AF**: LAF rises from 0.20 to 0.25 and HAF falls from 0.80 to 0.75, so the heterozygote allele-fraction interval **narrows** as W increases

Consistency metrics

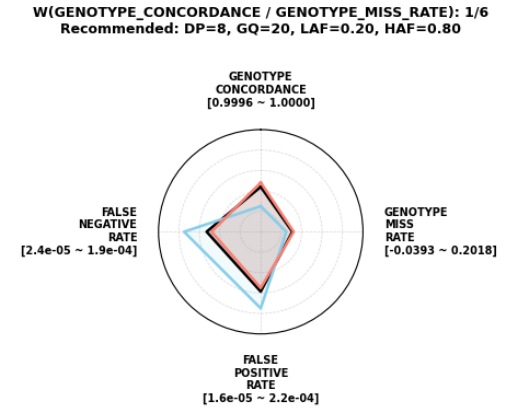
- **Concordance** rises from 99.975% to 99.984% (+0.009 percentage points)
- **Miss rate** rises from 3.534% to 13.232%, nearly 4 times higher
- The knee moves toward the **upper right** as W increases: the algorithm trades data volume for consistency

Beyond the low-W range, a 0.009-point concordance gain costs almost 10 percentage points more missing genotypes.

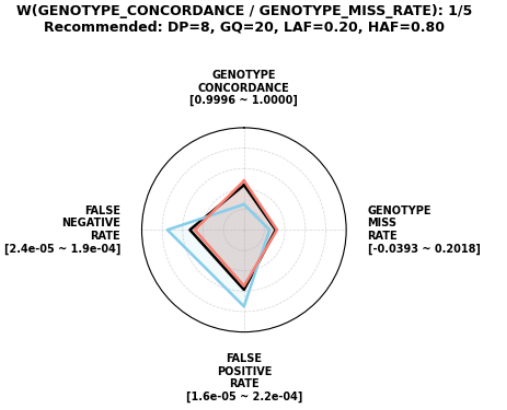
W = 1/7 to 1/4: One Setting, and 15x Misses More Genotypes



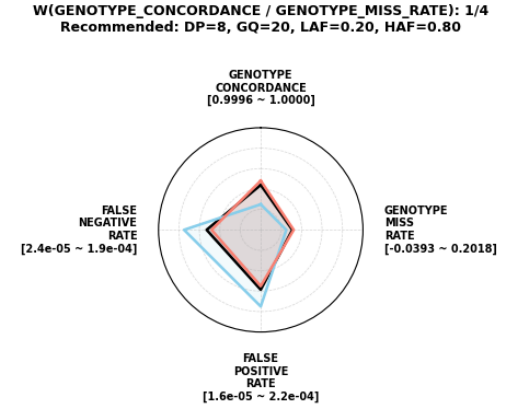
	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99975	0.03534	0.00013	0.00011
30X	0.99969	0.02095	0.00017	0.00015
15X	0.99977	0.03842	0.00013	0.00010



	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99975	0.03534	0.00013	0.00011
30X	0.99969	0.02095	0.00017	0.00015
15X	0.99977	0.03842	0.00013	0.00010



	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
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15X	0.99977	0.03842	0.00013	0.00010



	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99975	0.03534	0.00013	0.00011
30X	0.99969	0.02095	0.00017	0.00015
15X	0.99977	0.03842	0.00013	0.00010



Same knee for all four weights

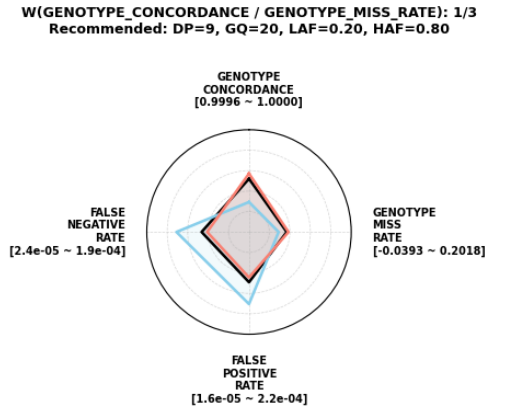
- DP 8, GQ 20, LAF 0.20, HAF 0.80
- All samples: concordance 99.975%, miss rate 3.53%

15x against 30x

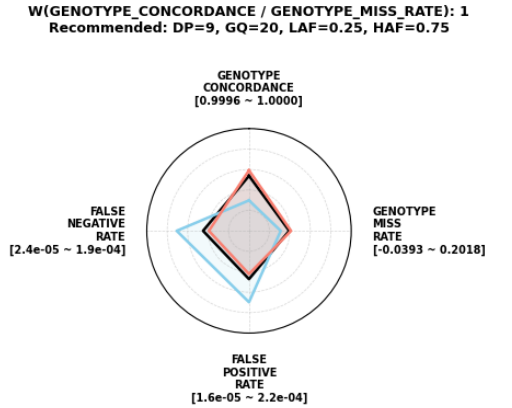
- Miss rate: **3.84%** at 15x against **2.10%** at 30x
- Concordance: 99.977% against 99.969%
- False positive and false negative rates: ≤ 0.017% on both

At the recommended setting the platforms agree on accuracy and differ only in missingness.

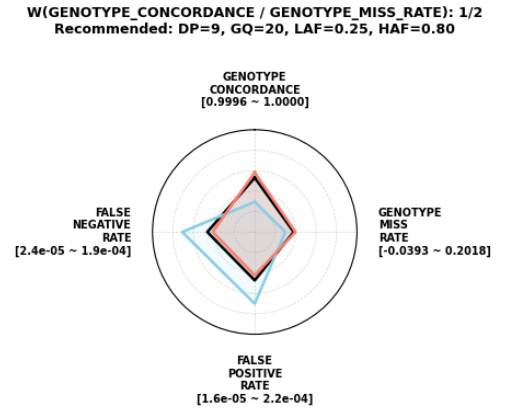
W = 1/3 to 2: Miss Rate Rises from 4.4% to 13.2%



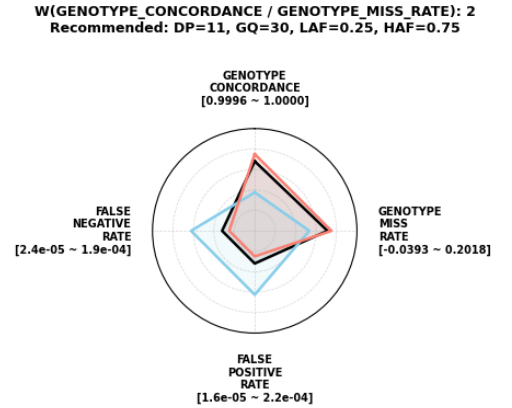
	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99978	0.04970	0.00011	0.00010
30X	0.99970	0.03053	0.00016	0.00014
15X	0.99980	0.05380	0.00010	0.00009



	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99979	0.05551	0.00011	0.00010
30X	0.99970	0.03548	0.00016	0.00014
15X	0.99981	0.05980	0.00010	0.00009



	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99979	0.05259	0.00011	0.00010
30X	0.99970	0.03298	0.00016	0.00014
15X	0.99981	0.05679	0.00010	0.00009



	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99984	0.13232	0.00008	0.00008
30X	0.99973	0.08943	0.00014	0.00013
15X	0.99987	0.14151	0.00007	0.00007



Knee moves to stricter settings

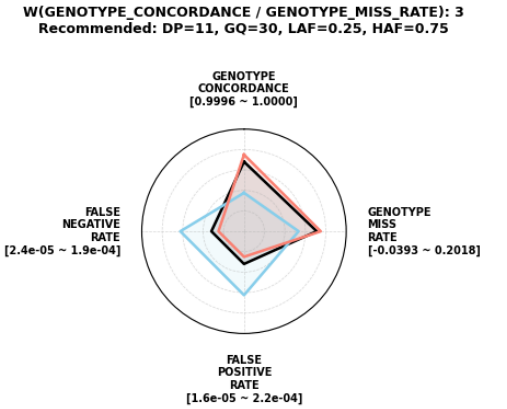
- W = 1/3: DP 9, GQ 20, LAF 0.20, HAF 0.80
- W = 1/2 and 1: DP 9, GQ 20, LAF 0.25 (HAF 0.80, then 0.75)
- W = 2: DP 11, GQ 30, LAF 0.25, HAF 0.75

All samples

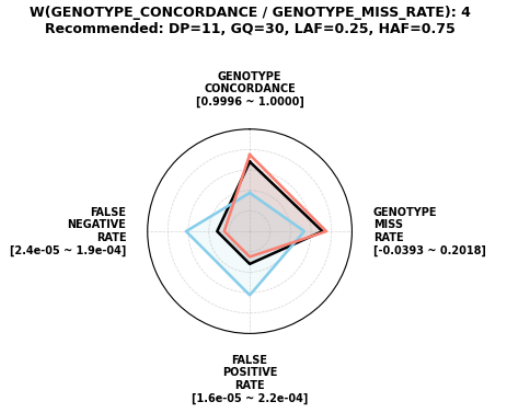
- Concordance: 99.978% → 99.984%
- Miss rate: 4.37% → 13.23%

From W = 1 to 2, a 0.005-point concordance gain more than doubles the miss rate.

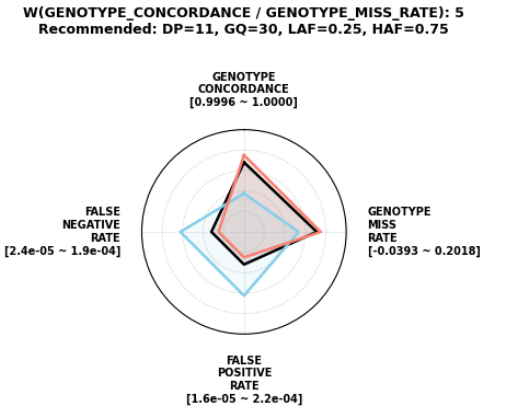
W = 3 to 6: Knee Stays at DP 11, GQ 30



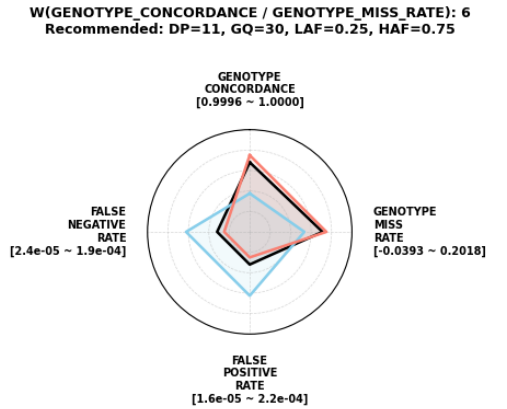
	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99984	0.13232	0.00008	0.00008
30X	0.99973	0.08943	0.00014	0.00013
15X	0.99987	0.14151	0.00007	0.00007



	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
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	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
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	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
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30X	0.99973	0.08943	0.00014	0.00013
15X	0.99987	0.14151	0.00007	0.00007



No further change past W = 2

- DP 11, GQ 30, LAF 0.25, HAF 0.75 for every weight
- All samples: concordance 99.984%, miss rate 13.23%

15x against 30x

- Miss rate: **14.15%** at 15x against **8.94%** at 30x
- Concordance: 99.987% against 99.973%

Strict settings widen the 15x–30x missingness gap to 5.2 points.

After Tuning, 15x and 30x Differ Mainly in Miss Rate

Across all weights tested (W = 1/7 to 6), comparing 15x, 30x and all samples.

Genotype concordance

- Remains **high for all three platforms** across all weight settings
- After global tuning, 15x and 30x show **highly comparable** concordance
- Cross-platform differences in concordance have been **substantially reduced**

Genotype miss rate

- Remains the **major residual difference** across platforms
- Under every setting, **15x consistently shows higher missingness** than 30x
- Depth-related missingness **cannot be fully eliminated** by parameter tuning alone

False positive & false negative rates

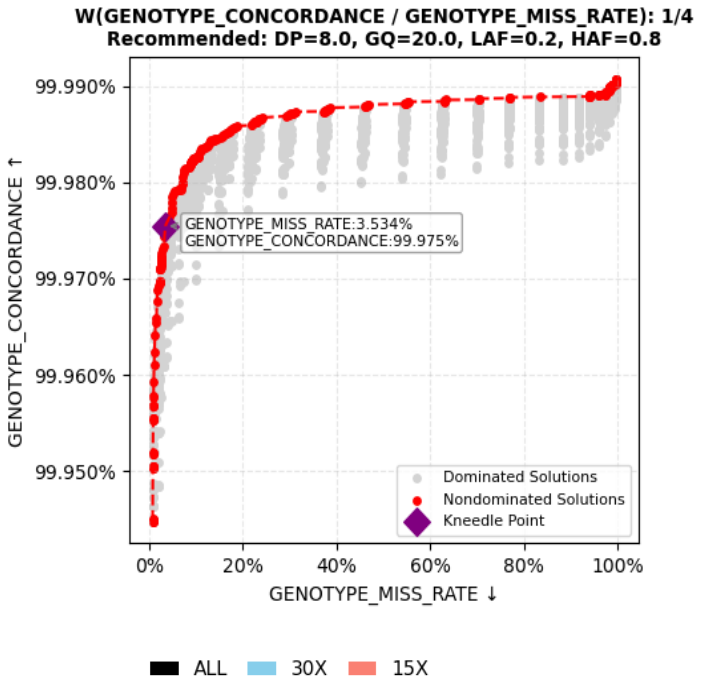
- After tuning, 15x and 30x are **broadly comparable** on both rates
- The remaining discrepancy is no longer driven by concordance or error rates
- It is mainly reflected in the **persistent imbalance of missingness**

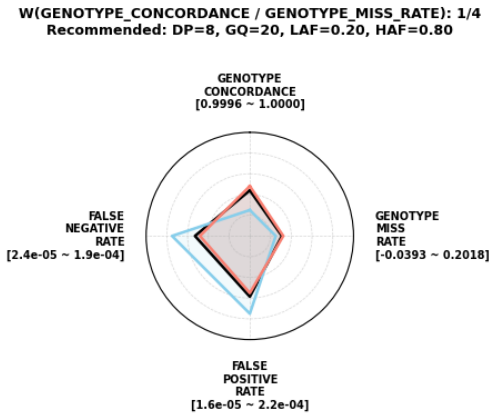
At the recommended setting (DP 8, GQ 20, LAF 0.20, HAF 0.80)

Platform	Concordance	Miss rate	False positive rate	False negative rate
All	99.975%	**3.53%**	0.013%	0.011%
30x	99.969%	**2.10%**	0.017%	0.015%
15x	99.977%	**3.84%**	0.013%	0.010%

Global tuning largely harmonized 15x and 30x on concordance and error rates; the remaining cross-platform gap is driven mainly by a persistent imbalance in genotype missingness.

Conclusion





	GENOTYPE CONCORDANCE	GENOTYPE MISS RATE	FALSE POSITIVE RATE	FALSE NEGATIVE RATE
ALL	0.99975	0.03534	0.00013	0.00011
30X	0.99969	0.02095	0.00017	0.00015
15X	0.99977	0.03842	0.00013	0.00010

Recommended: DP = 8, GQ = 20, LAF = 0.20, HAF = 0.80

Stable trade-off across a weight interval

- The same combination is selected for every W from 1/7 to 1/4, so it is **not specific to a single weighting scheme**
- Within this interval it stays on the Pareto front and is identified as the knee point, supporting its **robustness** as a practical trade-off

Balanced overall performance

- Concordance stays high (**99.975%**) while the miss rate stays moderate (**3.534%**): a reasonable balance between reliability and retention

Substantial cross-platform harmonization

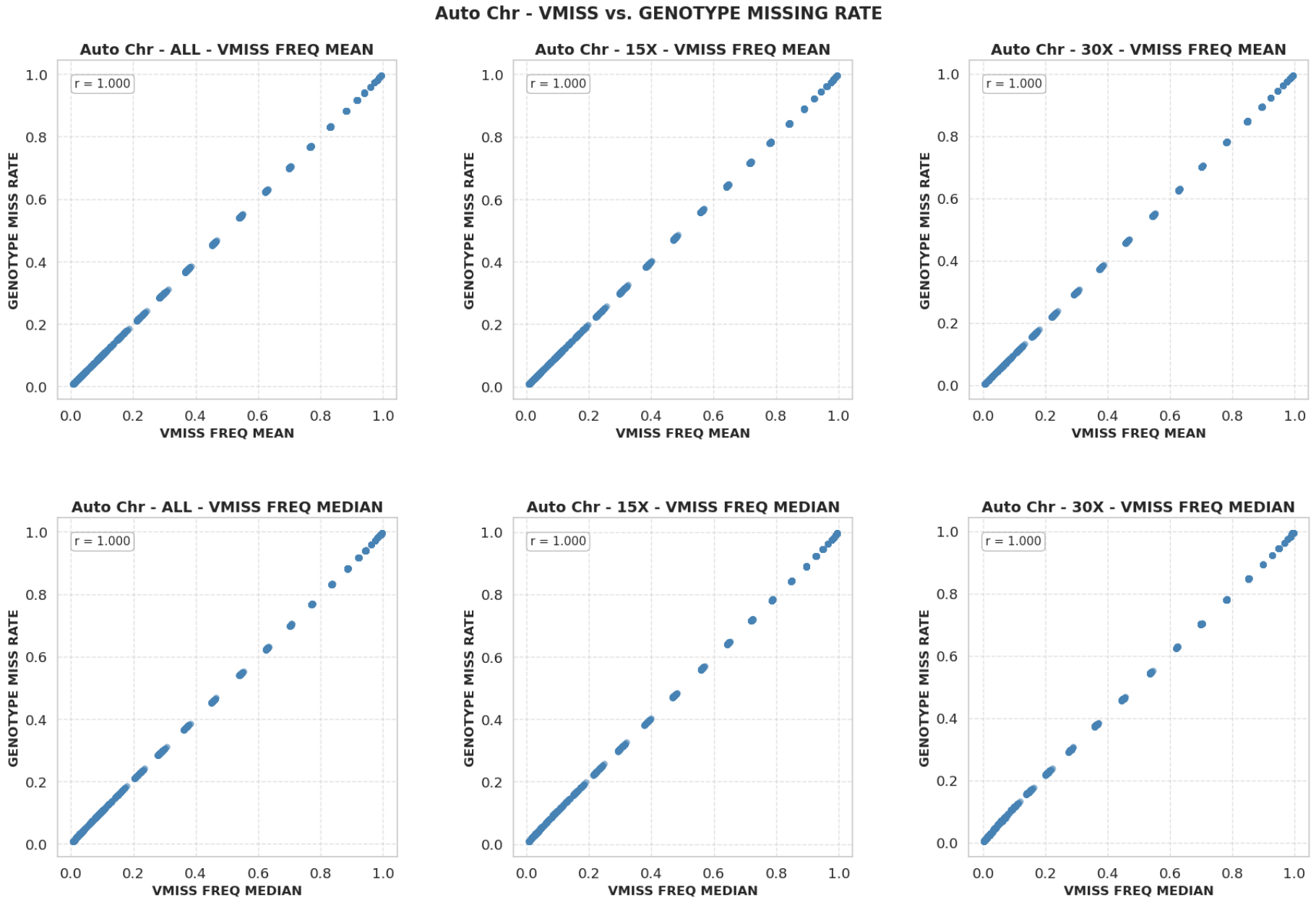
- 15x and 30x are highly comparable in concordance, false positive rate and false negative rate; global tuning reduces platform differences in the main accuracy metrics

Residual discrepancy driven by missingness

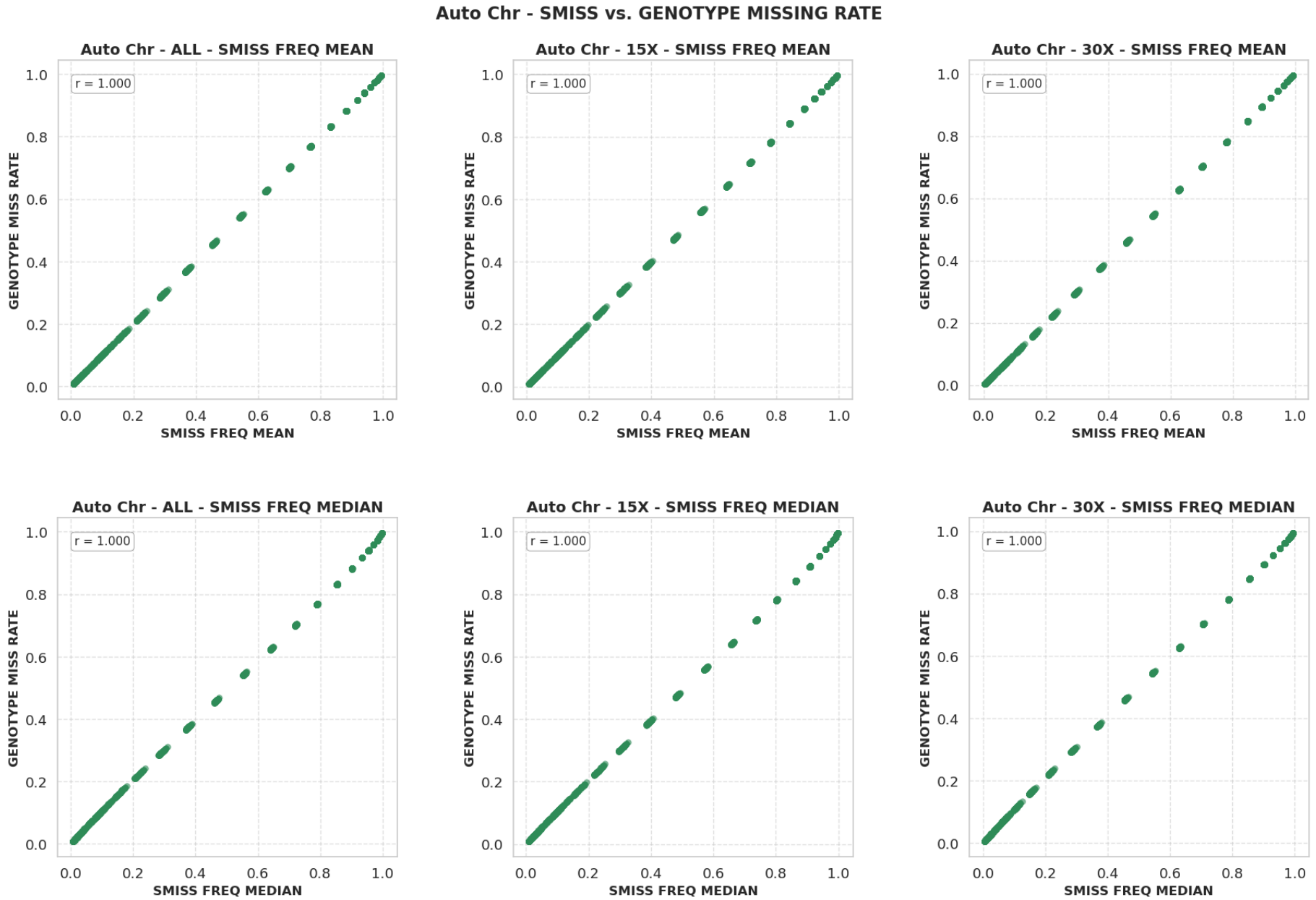
- 15x consistently shows **higher missingness** than 30x (3.84% vs 2.10%); depth-related missingness cannot be fully eliminated by tuning alone

Use DP 8, GQ 20, AF 0.20–0.80 as the default WGS genotype QC for this cohort.

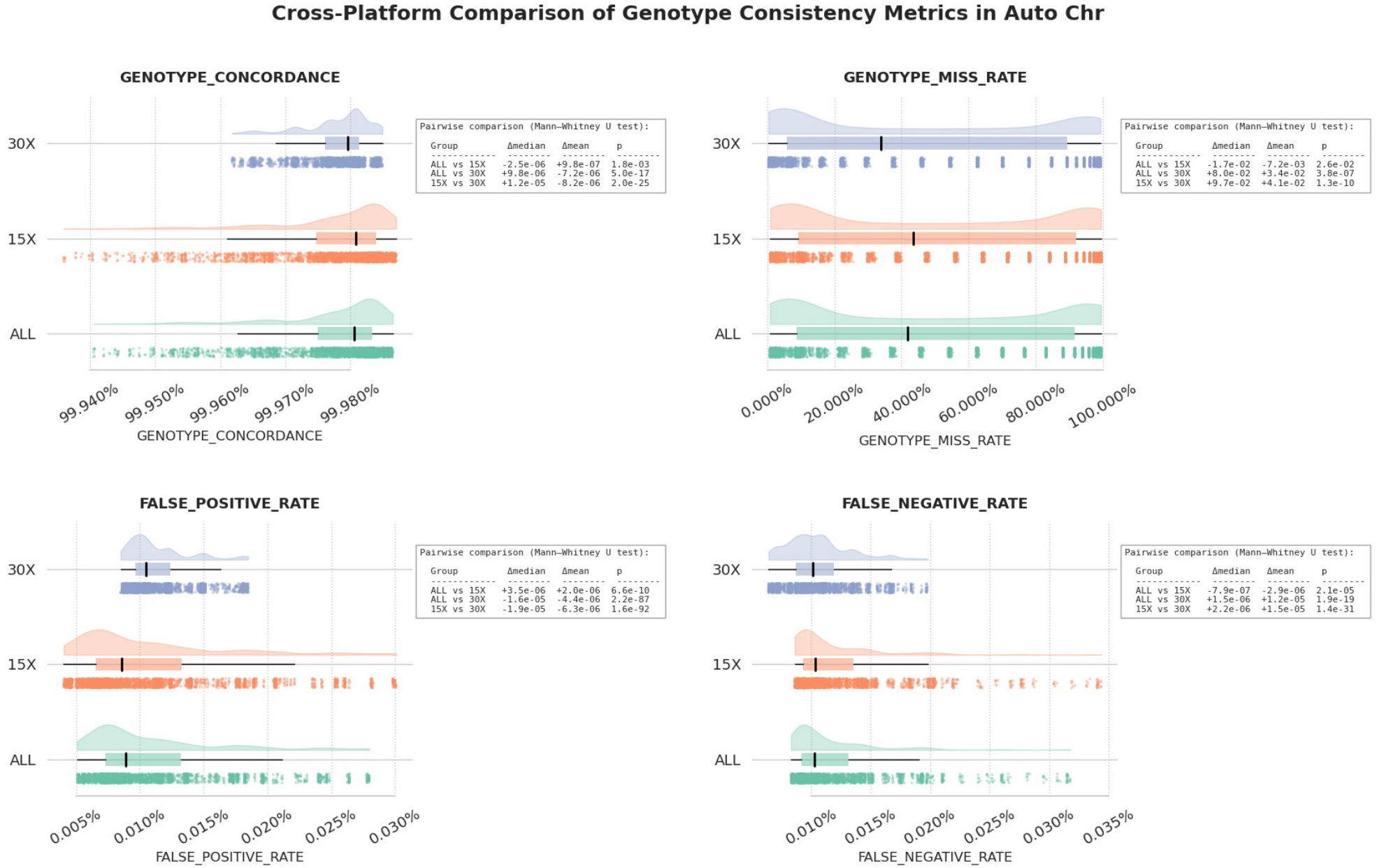
Genotype Miss Rate Tracks Variant Missingness (VMISS) Exactly, r = 1.000



Genotype Miss Rate Tracks Sample Missingness (SMISS) Exactly, r = 1.000



Consistency Metric Distributions by Platform (Autosomes)



Missingness at DP 8, GQ 20: 15x Averages 3.8%, 30x 2.1%

